Sampling from the Solution Space and Metabolic Environments of Genome-Scale Metabolic Models

October, 2025 @ School of Artificial Intelligence applied to microbiomea

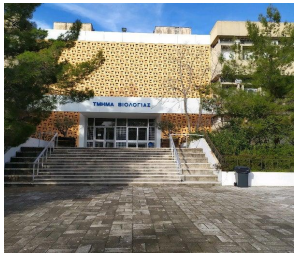
Haris Zafeiropoulos

hello friend





Ancient Olympia, Greece



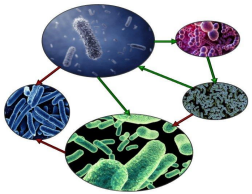
Athens, Greece
2010-2016







Heraklion, Greece
2016-2022





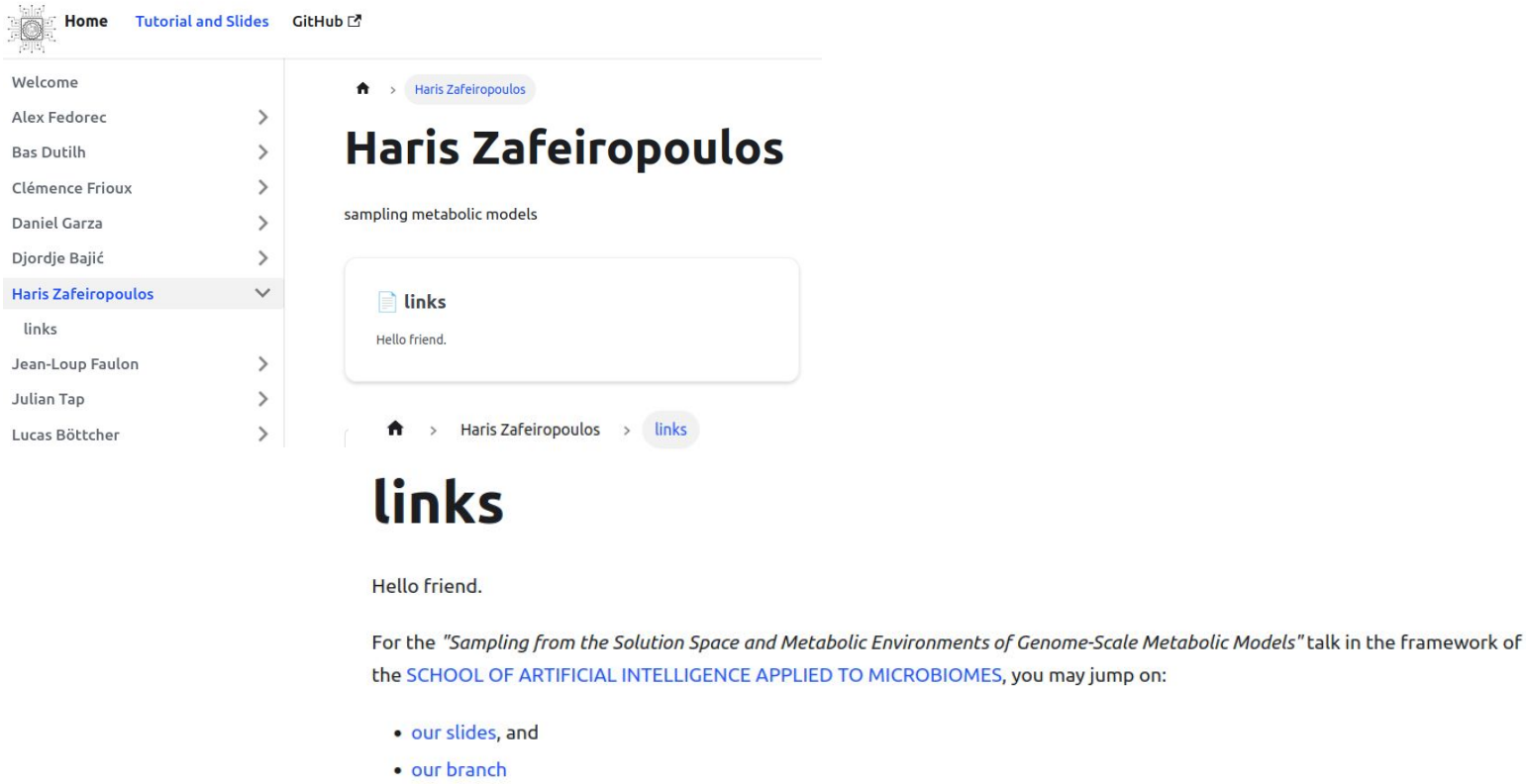
Leuven, Belgium
2022 - currently

hariszaf.github.io

github.com/hariszaf



Set the scene

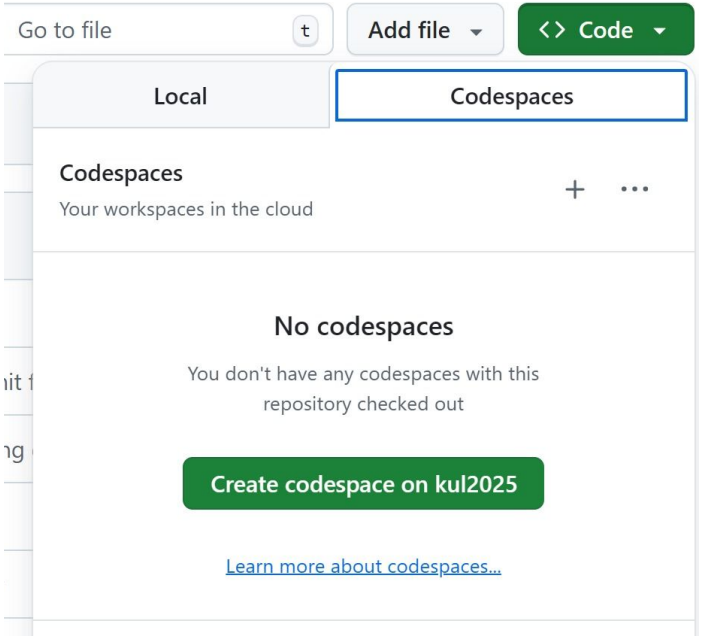

Home
Tutorial and Slides
GitHub
Welcome
Alex Fedorec
Bas Dutilh
Clémence Frioux
Daniel Garza
Djordje Bajić
Haris Zafeiropoulos
links
Jean-Loup Faulon
Julian Tap
Lucas Böttcher
Haris Zafeiropoulos
Haris Zafeiropoulos
sampling metabolic models
links
Hello friend.
Haris Zafeiropoulos
links
links
Hello friend.
For the "Sampling from the Solution Space and Metabolic Environments of Genome-Scale Metabolic Models" talk in the framework of the SCHOOL OF ARTIFICIAL INTELLIGENCE APPLIED TO MICROBIOMES, you may jump on:
our slides, and
our branch

Time to fire a codespace

1. (**Optional**) Make a fork of our `metabolic_toy_model` GitHub repository; take care you deselect the *Copy the master branch only* option. This gives you your own workspace where you can make changes, add new material, and experiment freely.

2. Make sure you make use of the ai-school25 branch, and

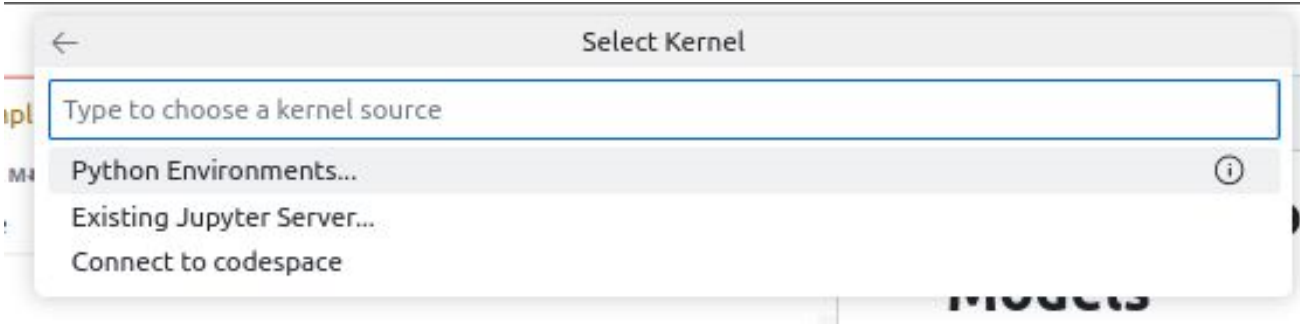
3. Click on the ***Code*** button, and then from the ***Codespaces*** tab, click on the ***Create codespace on ai-school25***



We will be following the following notebook, so we'll be switching between these slides and that

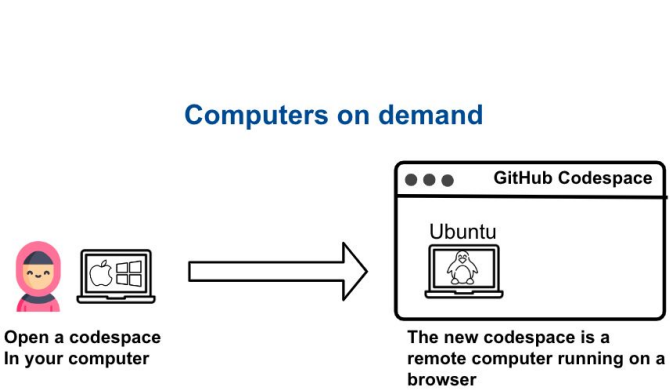
https://github.com/hariszaf/metabolic_toy_model/blob/ai-school25/flux_sampling.ipynb

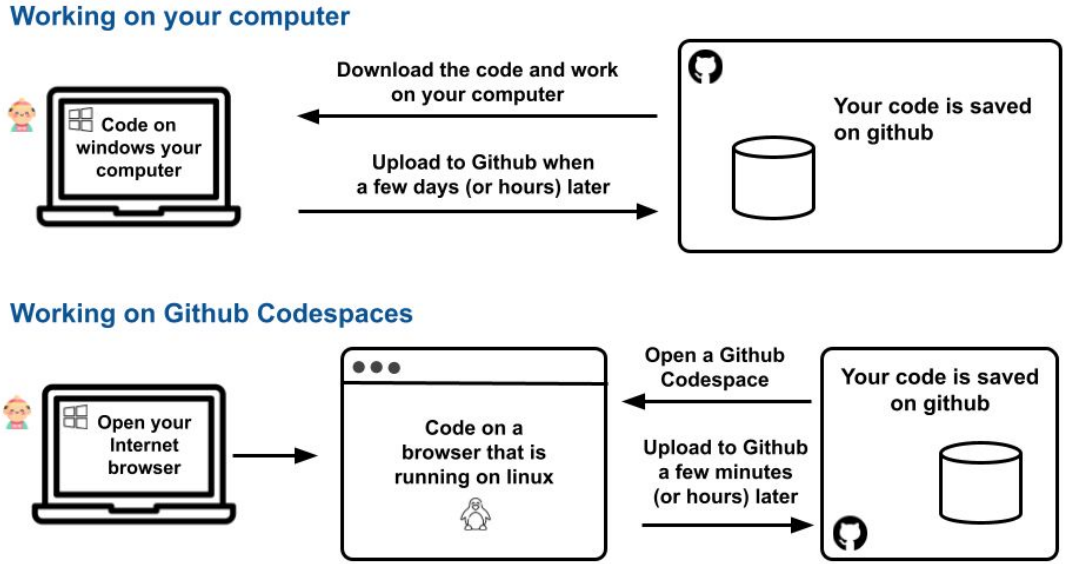
Try to run the first chunk



Click on Python environments, and then on the *global* (suggested) environment.

But.. what is a codespace?





https://github.com/codespaces/

The paradigm

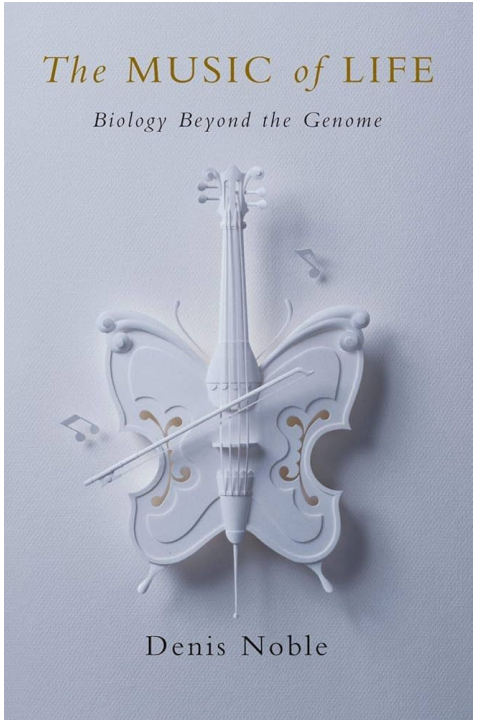
The burden of Reductionism

Traditionally, for studies relating **genetics** and **ecology** we **first** *identify* an

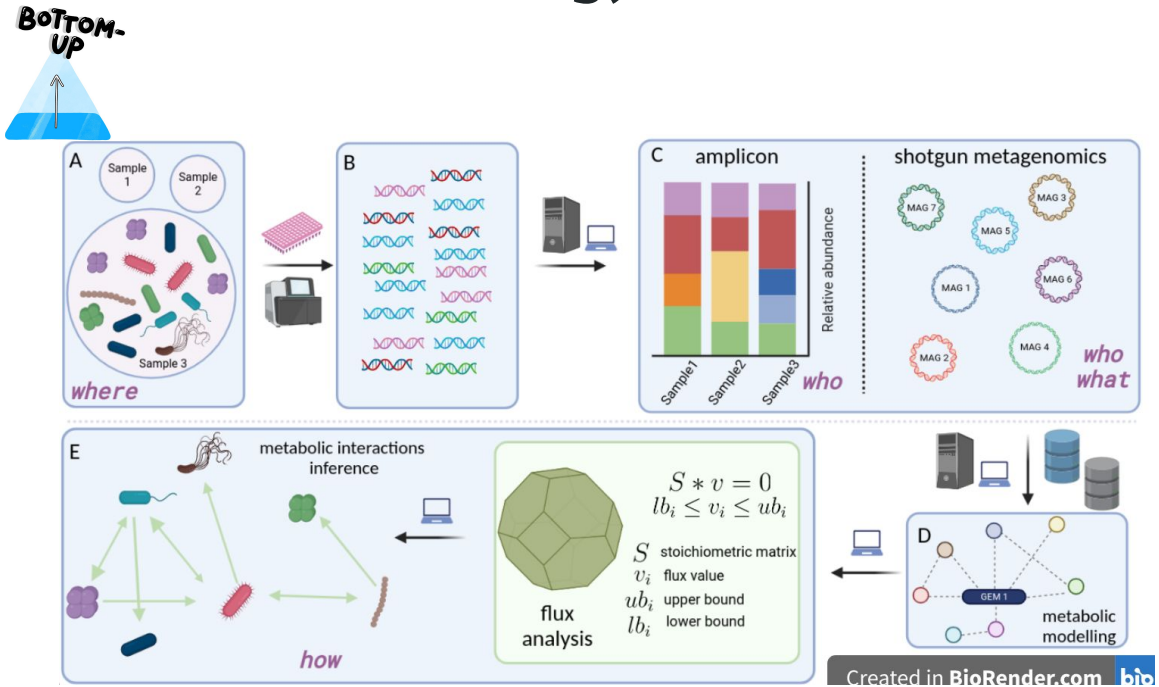
ecological adaptive phenotype

and **then** they try to *detect*

causal genetic variation



Reverse ecology



Among other tasks:

- ➔ describe growth medium
- ➔ predict microbe-microbe interactions
- ➔ identify pathways being used under different conditions
- ➔ key features of core metabolism and its evolution

Describe growth medium

Oxygen is required for growth, despite the presence of the metabolic mechanism for anaerobic respiration.

Although *Campylobacter* spp. can operate the electron transport chain in the absence of oxygen, **oxygen is nonetheless essential** for growth.

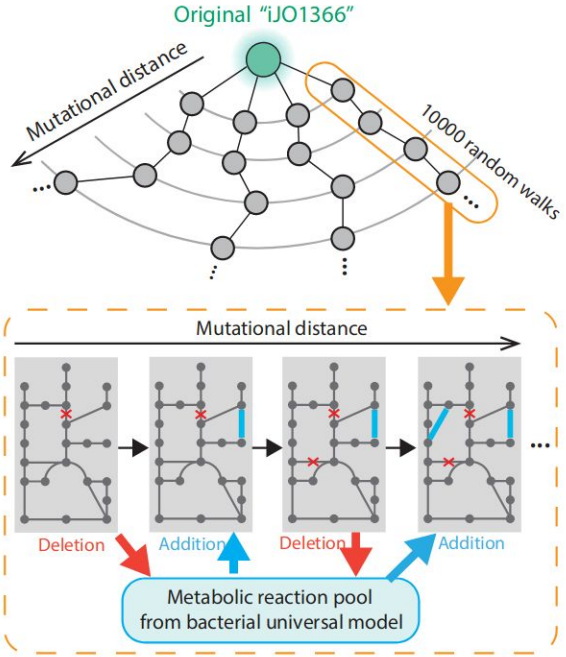
Substrates	DMEM/ F-12*	MM1	MM2	MM3	MM4
Auxotropic					
Methionine	115.7 μM	4 mM	4mM	8 mM	6.8 mM
Pantothenate	4.7 μM	9.4 μM	9.4 μM	9.4μM	9.4 μM
Niacinamide	16.6 μM	33.1 μM	33.1 μM	33.1 μM	33.1 μM
Carbon source					
Pyruvate	0.5 mM	10 mM	10 mM	10 mM	10 mM
Growth-improving					
Cysteine·HCl	100 μM	4 mM	4 mM	–	4 mM
Serine	250 μM	10 mM	10 mM	10 mM	–
Glutamine	2.5 mM	10 mM	–	10 mM	13.6
Glutamate	50 μM	–	20 mM	–	–
Inorganic salts	same as DMEM/F-12 (**Table S1**)				
Growth (OD 600)					
24 h	0.38 ± 0.01	0.31 ± 0.05	0.14 ± 0.03	0.02 ± 0.01	0.15 ± 0.01
48 h	0.34 ± 0.04	1.05 ± 0.06	1.07 ± 0.09	0.41 ± 0.04	0.57 ± 0.01

OD results expressed as mean ± stdev (n = 3).
**Table S1 for all the components of DMEM/F-12.*

Figure from:
Tejera N, Crossman L, Pearson B, Stoakes E, Nasher F, Djeghout B, Poolman M, Wain J, Singh D. Genome-scale metabolic model driven design of a defined medium for Campylobacter jejuni M1cam. Frontiers in Microbiology. 2020 Jun 19;11:1072.

Key features of core metabolism and its evolution

Some sugar hierarchies are easier to evolve than others.



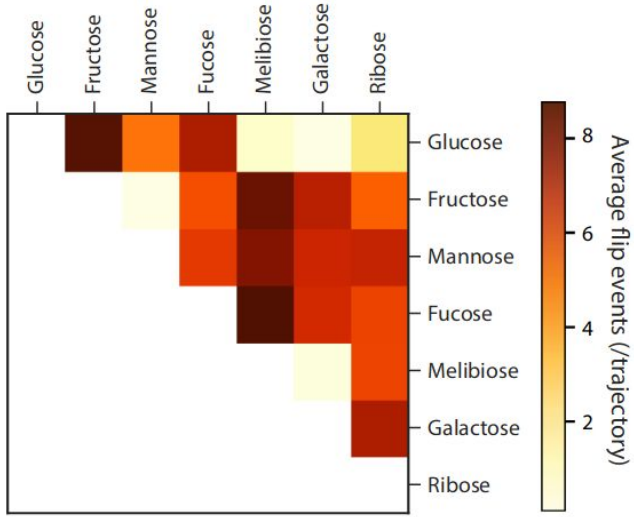


Figure from:
Takano S, Vila JC, Miyazaki R, Sánchez Á, Bajić D. The architecture of metabolic networks constrains the evolution of microbial resource hierarchies. Molecular biology and evolution. 2023 Sep;40(9):msad187.

Other applications ..?

13:30 - 15:30	Metabolic network reconstruction and modelling - from the individual to the community	Clémence Frioux

More applications ..?

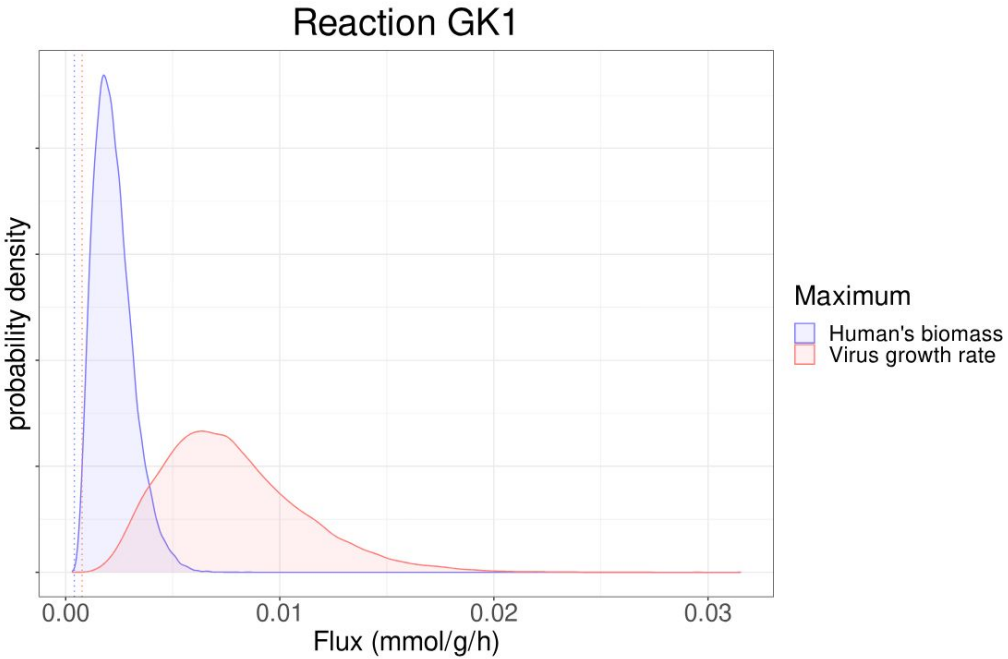


Figure from:
Our blog post on tweag

This is our endgame for today.

But what are we looking at and how is this different than what we have discussed so far?

The model

metabolic models: recap

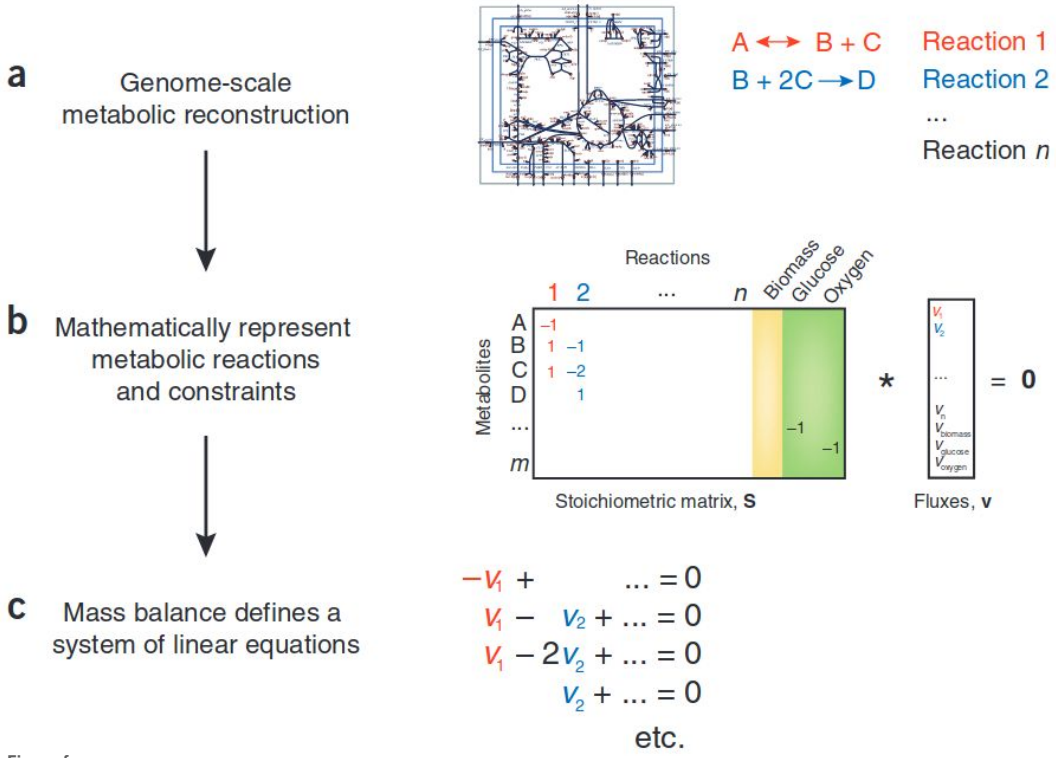
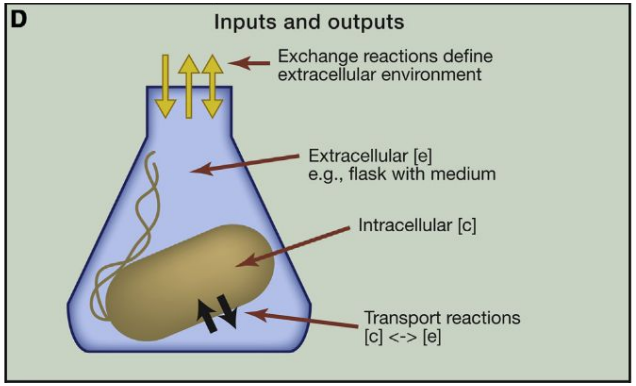


Figure from:
Orth JD, Thiele I, Palsson BØ. What is flux balance analysis?. Nature biotechnology. 2010 Mar;28(3):245-8.



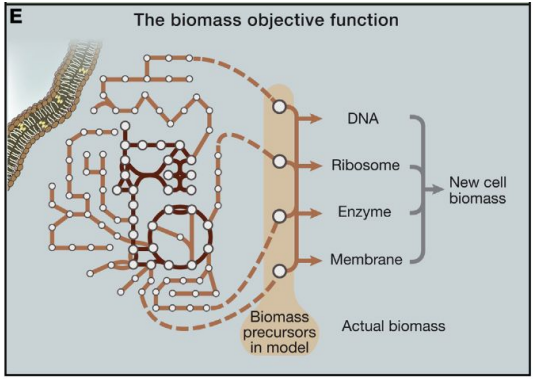
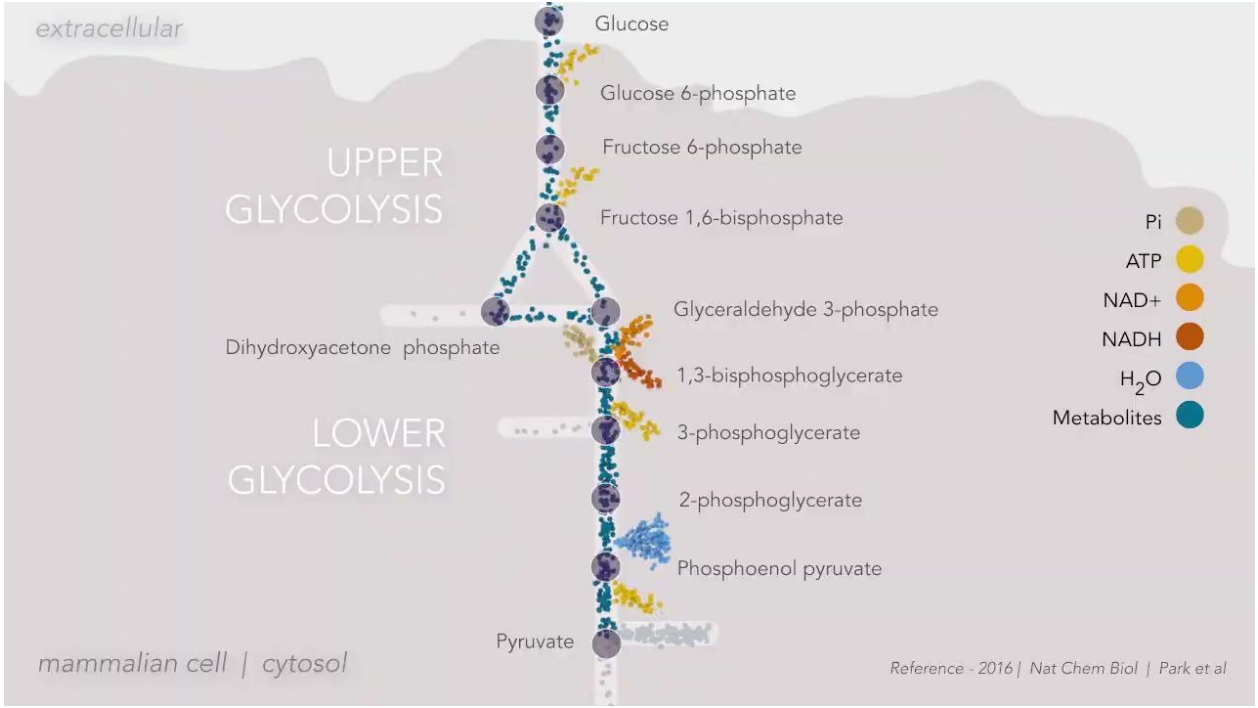


Figure from:
O'Brien EJ, Monk JM, Palsson BO. Using genome-scale models to predict biological capabilities. Cell. 2015 May 21;161(5):971-87.

The rate at which a reaction proceeds within a metabolic network



Video from:
https://iwasa.biochem.utah.edu/metabolism/index.html

mmol/(gDW·h) (millimoles per gram of dry weight per hour)

A (somehow different) phylogeny

Biased vs. Unbiased

What splits these two branches is the presence of an objective function

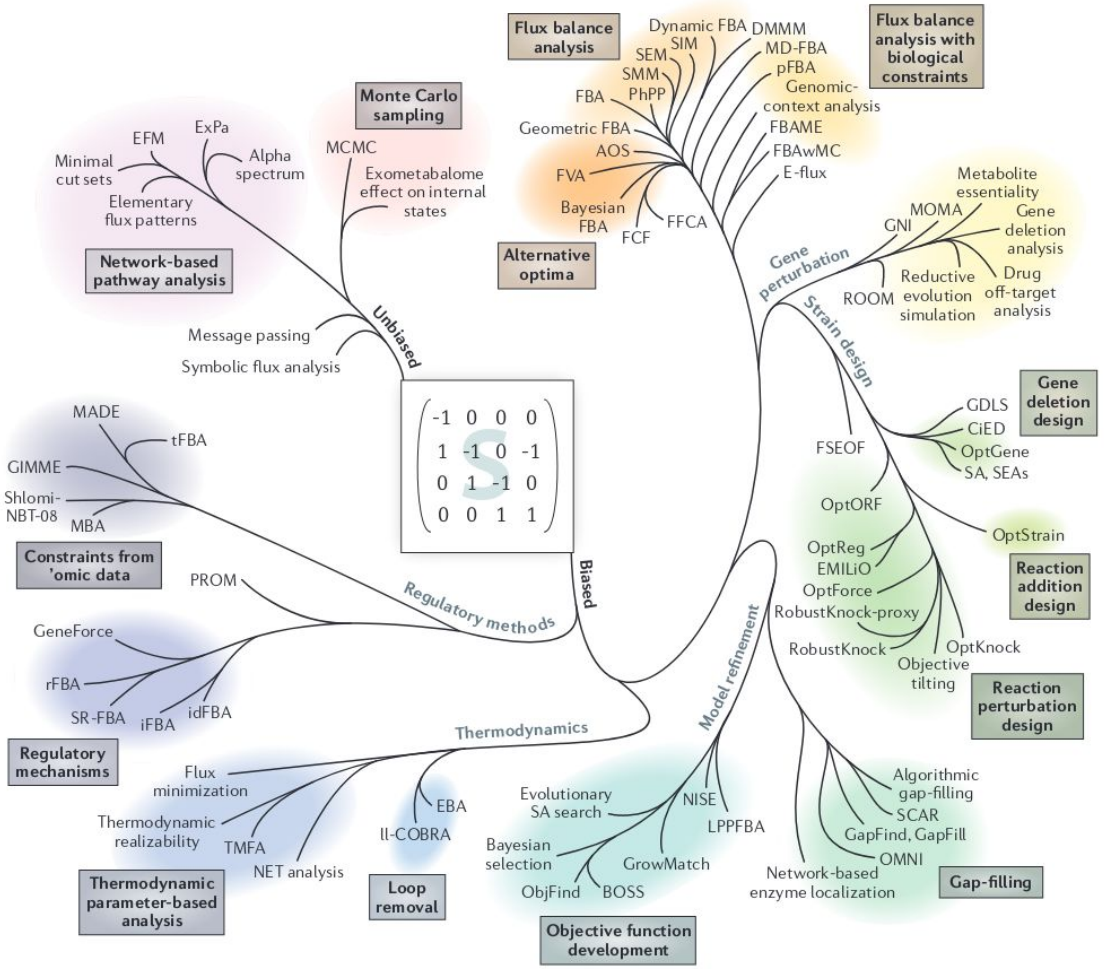
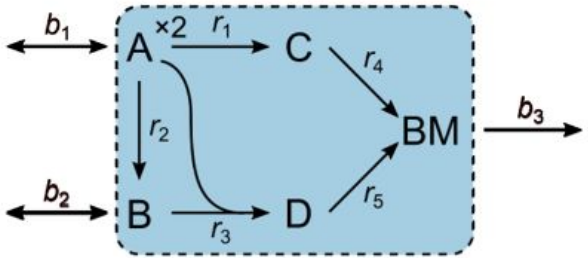


Figure from:
Lewis NE, Nagarajan H, Palsson BO. Constraining the metabolic genotype–phenotype relationship using a phylogeny of in silico methods. Nature Reviews Microbiology. 2012 Apr;10(4):291-305.

Optimization vs. enumeration

A

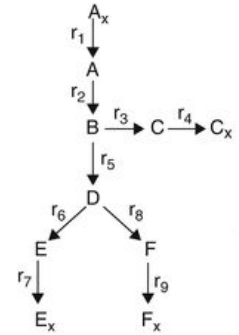


B

	ECM1	ECM2	ECM3	EFP1	EFP2	EFP3	MP1
A	-1	-2	-1	-	-	-	-
B	-1		+1	-		+	
BM	+1	+1		+	+		+

Figure from:
Wedmark YK, Vik JO, Øyås O. A hierarchy of metabolite exchanges in metabolic models of microbial species and communities. PLOS Computational Biology. 2024 Sep 26;20(9):e1012472.

a Network



b Mass Balance equations

$$S.r = 0$$

$$\frac{dA}{dt} = r_1 - r_2 = 0$$

$$\frac{dB}{dt} = r_2 - r_3 - r_5 = 0$$

$$\frac{dC}{dt} = r_3 - r_4 = 0$$

$$\frac{dD}{dt} = r_5 - r_6 - r_8 = 0$$

$$\frac{dE}{dt} = r_6 - r_7 = 0$$

$$\frac{dF}{dt} = r_8 - r_9 = 0$$

c Stoichiometric Matrix

	r_1	r_2	r_3	r_4	r_5	r_6	r_7	r_8	r_9
A	1	−1	0	0	0	0	0	0	0
B	0	1	−1	0	−1	0	0	0	0
C	0	0	1	−1	0	0	0	0	0
D	0	0	0	0	1	−1	0	−1	0
E	0	0	0	0	0	1	−1	0	0
F	0	0	0	0	0	0	0	1	−1

d Constraints

Objective function = Maximize (r_7)

$$0 \le r_1 < 1$$

$$0.2 \le r_4 \le 1$$

$$0.3 \le r_9 \le 1$$

e Solution

$$r = \begin{bmatrix} r_1 \\ r_2 \\ r_3 \\ r_4 \\ r_5 \\ r_6 \\ r_7 \\ r_8 \\ r_9 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0.2 \\ 0.2 \\ 0.8 \\ 0.5 \\ 0.5 \\ 0.3 \\ 0.3 \end{bmatrix}$$

Figure from:
Rajvanshi M, Venkatesh KV. Flux balance analysis. InEncyclopedia of systems biology 2013 (pp. 749-752). Springer, New York, NY.

Why a certain solution is optimal?

Elementary Flyx Modes (EFMs) can be viewed as the **minimal metabolic strategies** that a cell has at its disposal, and *each metabolic state is a combination of these* EFMs.

Therefore, **understanding FBA** solutions comes down to **rationalizing why the selected EFMs are used**.

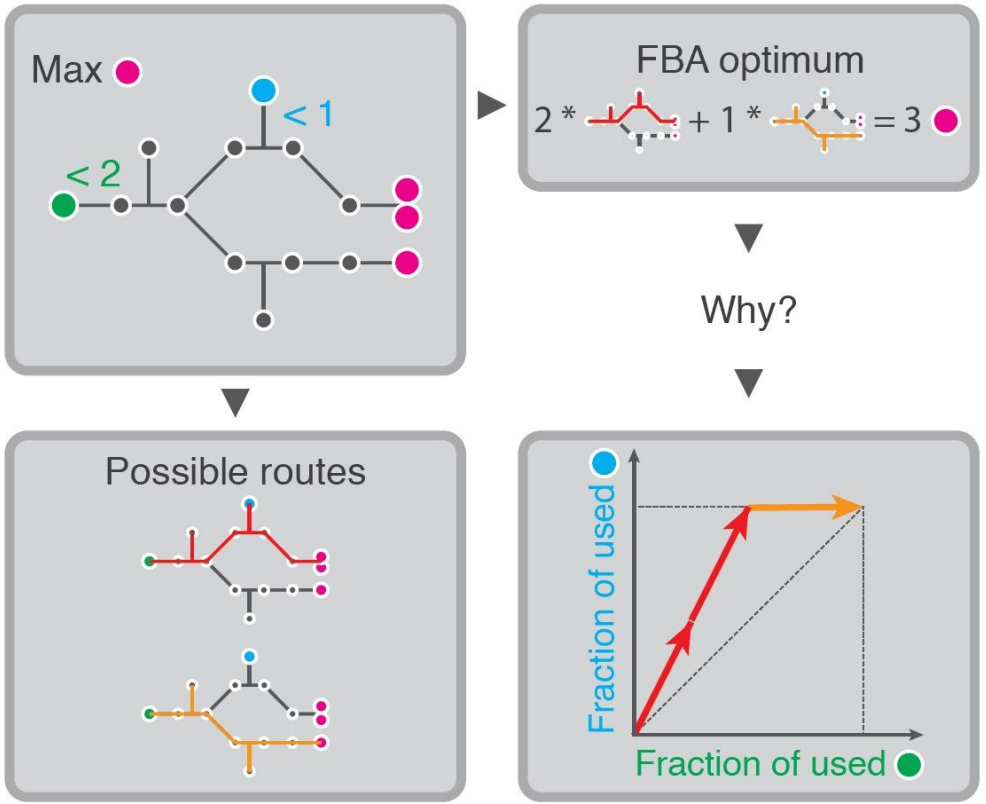
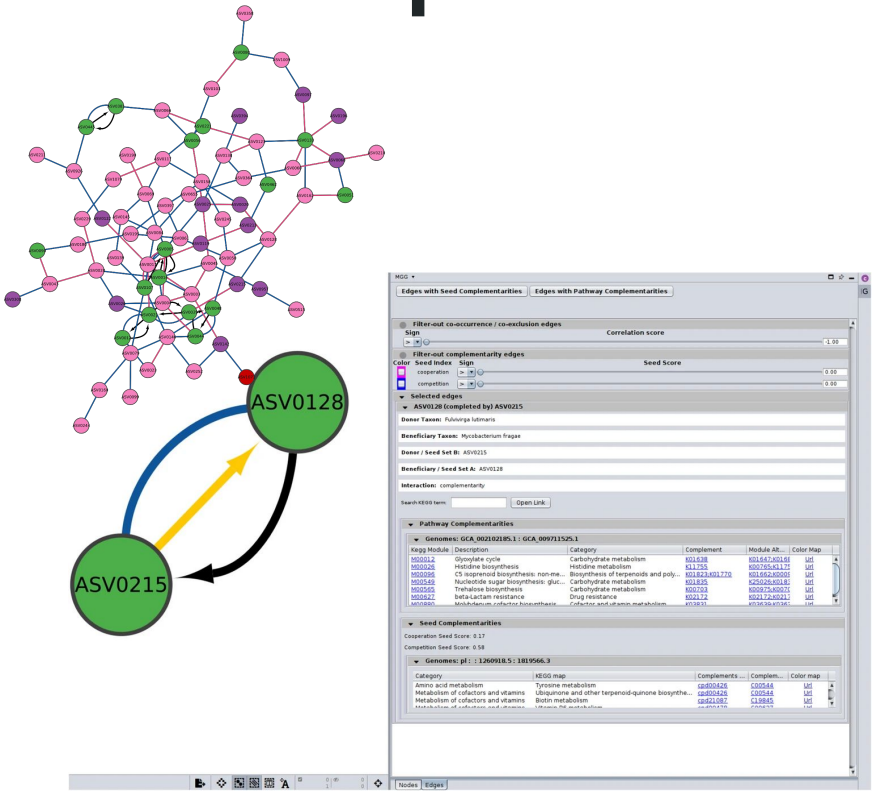
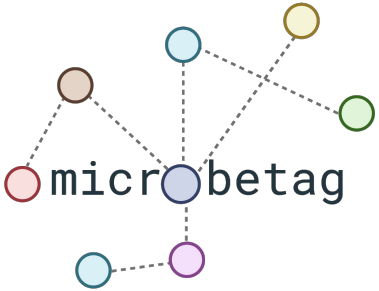
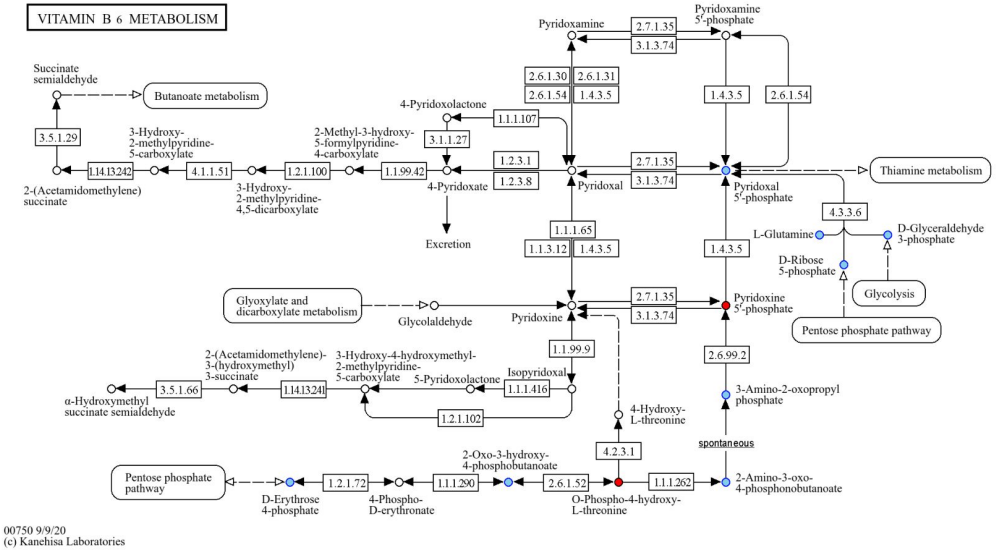


Figure from:
van Pelt-KleinJan E, De Groot DH, Teusink B. Understanding FBA solutions under multiple nutrient limitations. Metabolites. 2021 Apr 21;11(5):257.

A short parenthesis..







.. to annotate some co-occurence networks

Zafeiropoulos *et al. Genome Biology* (2025) 26:292
https://doi.org/10.1186/s13059-025-03769-2

Genome Biology

SOFTWARE Open Access

microbetag: simplifying microbial network interpretation through annotation, enrichment tests, and metabolic complementarity analysis

Haris Zafeiropoulos[1*], Ermis Ioannis Michail Delopoulos[1], Andi Erega[2], Aline Schneider[2], Annelies Geirnaert[2], John Morris[3] and Karoline Faust[1*]

github.com/hariszaf/metabolic_toy_model/tree/kul2025

microbetag.readthedocs.io

matrix.to/#/#microbetagcommunity:matrix.org

github.com/msysbio/microbetag

youtube.com/watch?v=pLMK1yVv20M

doi.org/10.1186/s13059-025-03769-2

The challenge

Flux Balance Analysis (FBA) limitations

- ➔ a **biased** method need for an (objective function)
- ➔ solution **uniqueness**
- ➔ **suboptimality**



FBA: recap

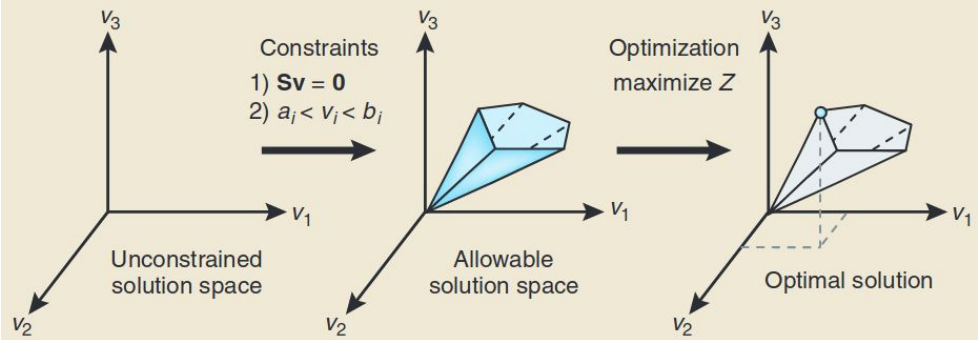


Figure 1 The conceptual basis of constraint-based modeling. With no constraints, the flux distribution of a biological network may lie at any point in a solution space. When mass balance constraints imposed by the stoichiometric matrix **S** (labeled 1) and capacity constraints imposed by the lower and upper bounds (a_i and b_i) (labeled 2) are applied to a network, it defines an allowable solution space. The network may acquire any flux distribution within this space, but points outside this space are denied by the constraints. Through optimization of an objective function, FBA can identify a single optimal flux distribution that lies on the edge of the allowable solution space.

Figure from:
Orth JD, Thiele I, Palsson BØ. What is flux balance analysis?. Nature biotechnology. 2010 Mar;28(3):245-8.

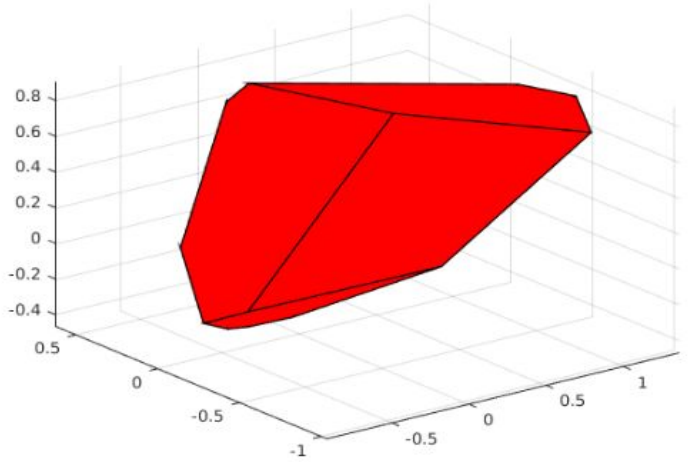
Recap in today's first talk

$$Sv = dx/dt$$

We assume that the net production and consumption of each and every metabolite within the cell is 0.

$$Sv = 0$$

Considering the system at steady state



Solution uniqueness

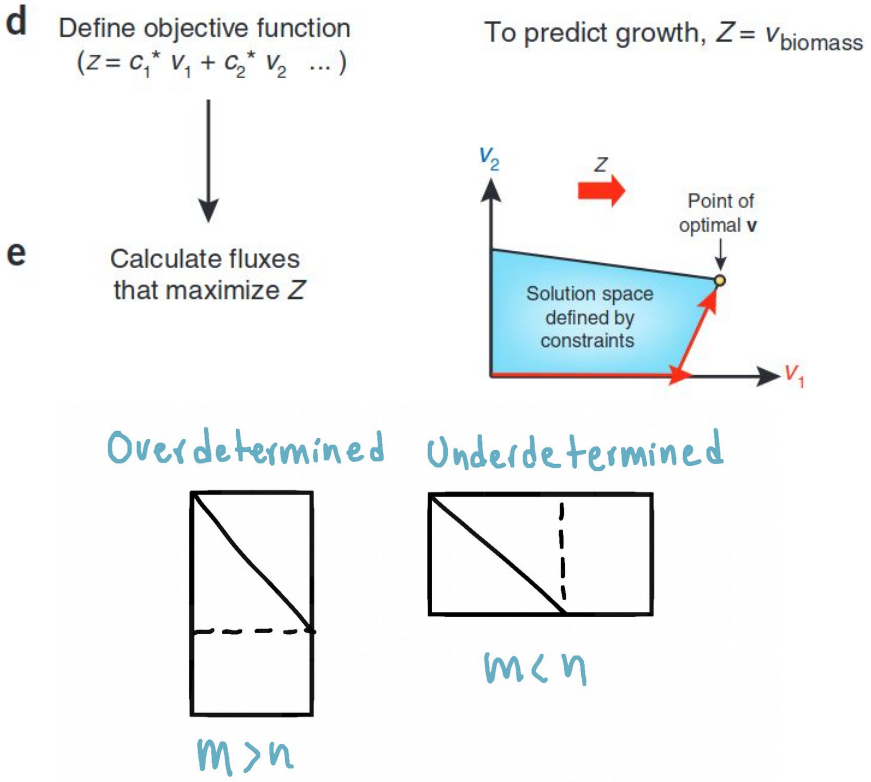
Undetermined problem

Variables = the **reaction fluxes** (v_1,v_2,...,v_n).

Equations = the **mass-balance (steady-state) constraints**, i.e. rows of the **stoichiometric matrix S**.

m: number of metabolites

n: number of reactions



Let's see an example

Toy Metabolic Model

v_1: $\mathrm{EX}_A \rightarrow A$ $(0 \leq v_1 \leq 10)$
v_2: $A \rightarrow B$
v_3: $A \rightarrow C$
v_4: $B \rightarrow \mathrm{Biomass}$
v_5: $C \rightarrow \mathrm{Biomass}$

$$S = \begin{bmatrix} 1 & -1 & -1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{bmatrix}, \; Sv = 0$$

Steady-state constraints:

$$\begin{cases} v_1 - v_2 - v_3 = 0 \\ v_2 - v_4 = 0 \\ v_3 - v_5 = 0 \end{cases}$$

Objective (maximize biomass):

$$\max Z = v_4 + v_5$$

Substitution:

$$\begin{aligned} v_4 &= v_2 \\ v_5 &= v_3 \\ v_1 &= v_2 + v_3 \end{aligned}$$

$$\Rightarrow \quad Z = v_2 + v_3 = v_1$$

Constraint:

$$v_1 \leq 10 \Rightarrow Z^\star = 10$$

Family of optimal solutions:

$$(v_1, v_2, v_3, v_4, v_5) = (10,\ t,\ 10-t,\ t,\ 10-t),\ t \in [0,10]$$

Examples:

$$t = 10 : (10,10,0,10,0)$$
$$t = 0 : (10,0,10,0,10)$$
$$t = 5 : (10,5,5,5,5)$$

- FBA will return a single flux vector

- different *solvers*, will return different vectors -- bye, bye reproducibility

Suboptimality

Organisms might not be operating at maximum capacity

(Oberhardt MA, Palsson BØ, Papin JA. Applications of genome-scale metabolic reconstructions. Mol Syst Biol. 2009; 5:320.).

Metabolic systems may not necessarily operate at the full optimal state, but may be operating in near optimal or sub-optimal modes

(Mahadevan R, Schilling CH. The effects of alternate optimal solutions in constraint-based genome-scale metabolic models. Metabolic engineering. 2003 Oct 1;5(4):264-76.)

In this case, the objective function might not be fully optimized, but instead be in a near-optimal or sub-optimal state.

(Schultz A, Qutub AA. Predicting internal cell fluxes at sub-optimal growth. BMC systems biology. 2015 Apr 3;9(1):18.)

Flux Sampling intro & approaches

Uniform random flux sampling

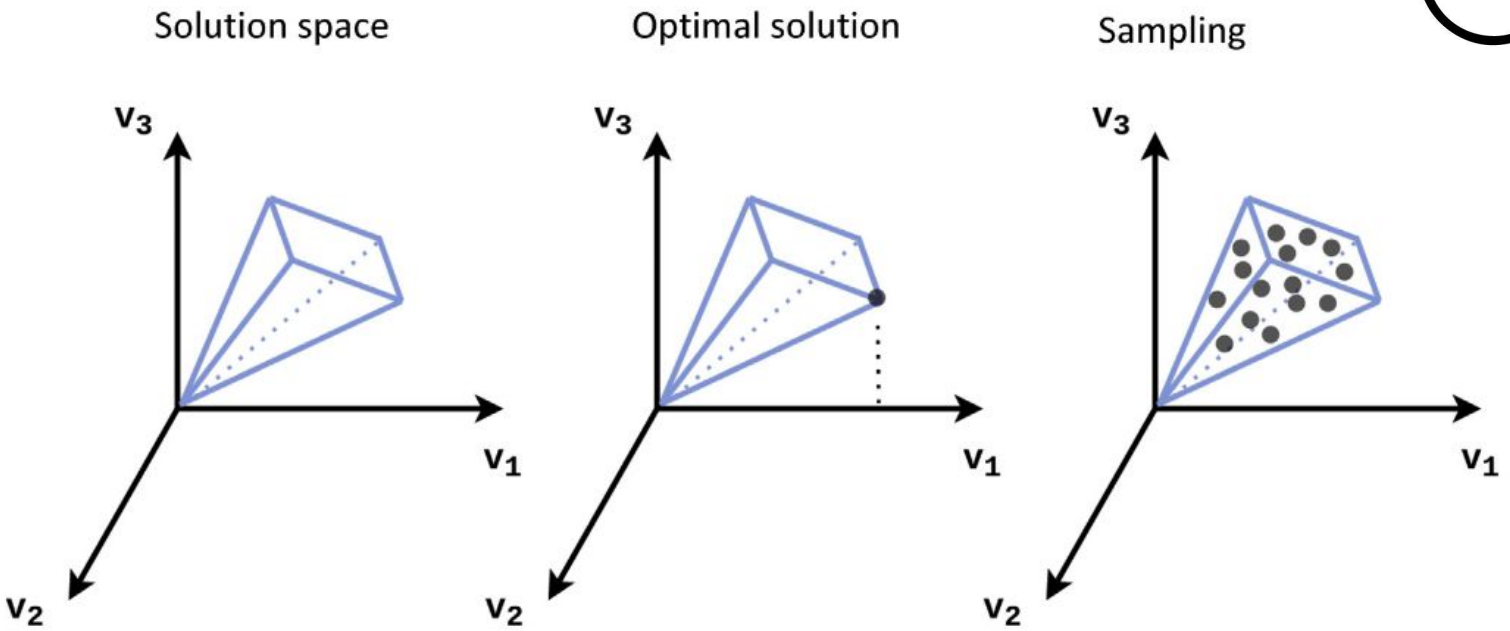


Figure from:
Garza DR, Gonze D, Zafeiropoulos H, Liu B, Faust K. Metabolic models of human gut microbiota: Advances and challenges. Cell systems. 2023 Feb 15;14(2):109-21

Constraints, (sub) optimal values, FBA and sampling

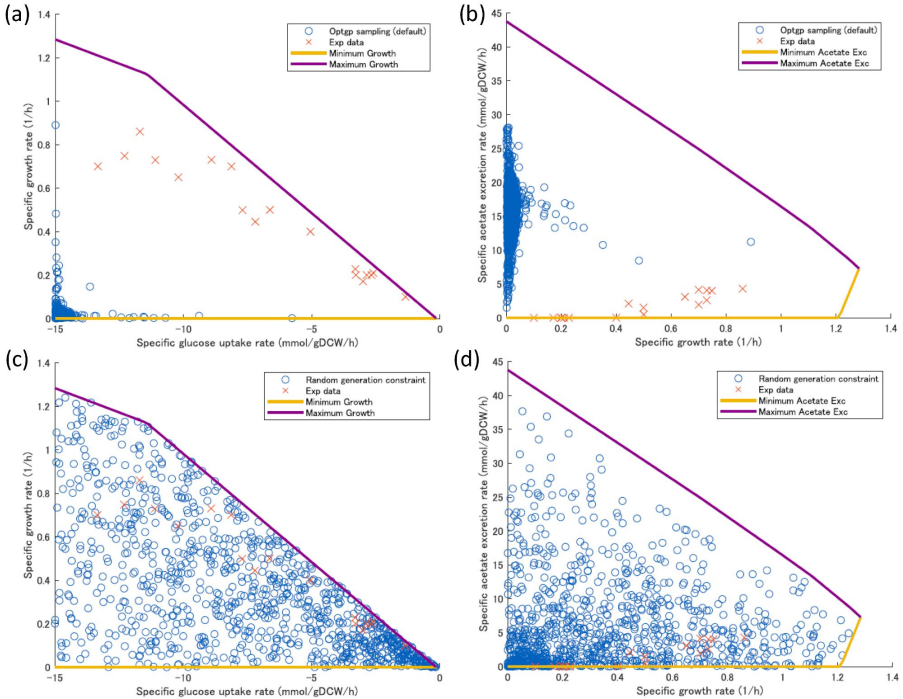
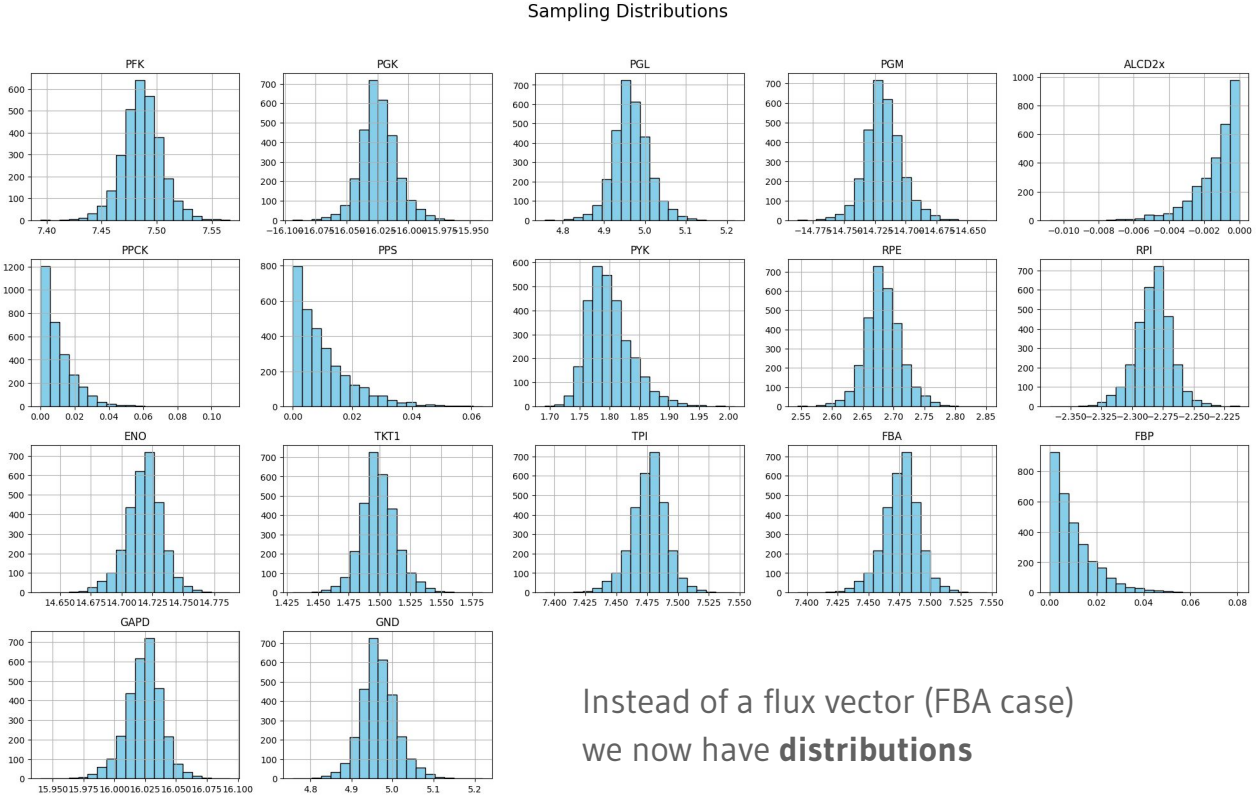


Figure from:
Kuriya Y, Murata M, Yamamoto M, Watanabe N, Araki M. Prediction of Metabolic Flux Distribution by Flux Sampling: As a Case Study, Acetate Production from Glucose in Escherichia coli. Bioengineering. 2023 May 24;10(6):636.

Final output



Instead of a flux vector (FBA case)
we now have **distributions**

What is a polytope ?

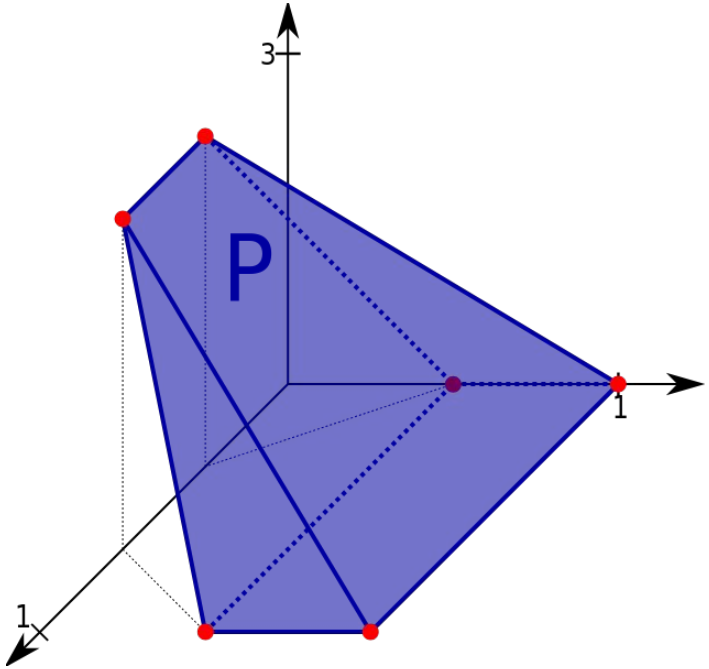
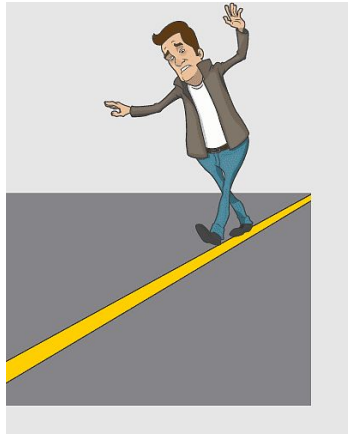
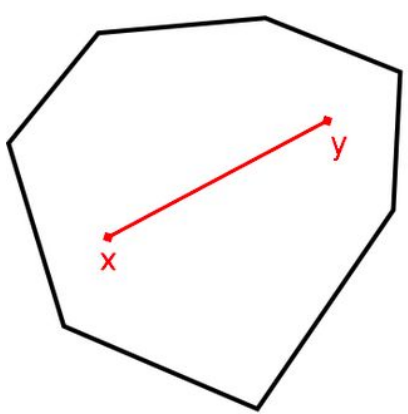


Figure from:
Wikie: https://en.wikipedia.org/wiki/Convex_polytope

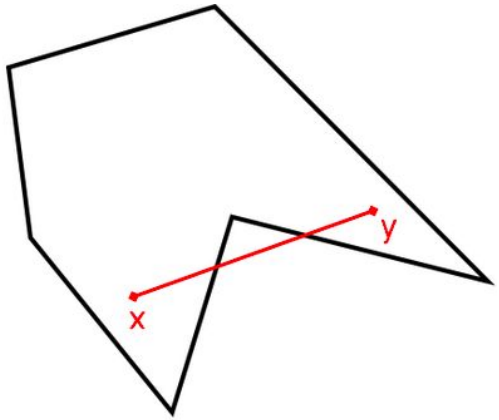
A polytope is called *full-dimensional* if it is an n-dimensional object in $\mathbb{R}^n$

Can you go from one place to the other following a straight line?





Convex region



Non-convex region

Figure from:
Katsilieris F. Search and secure using mobile robots.

Thermodynamic Infeasible solutions troubles!

Let’s go for a random walk..

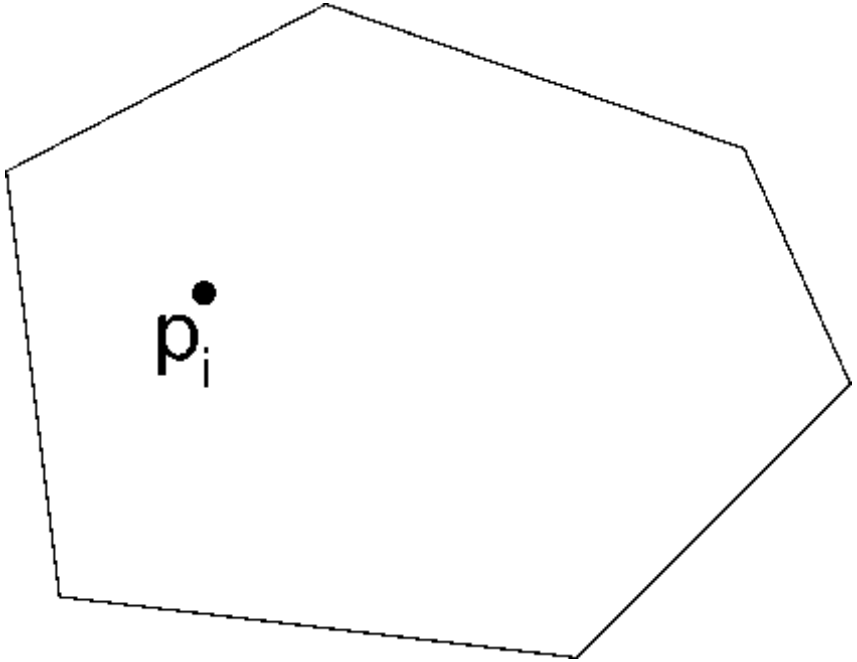
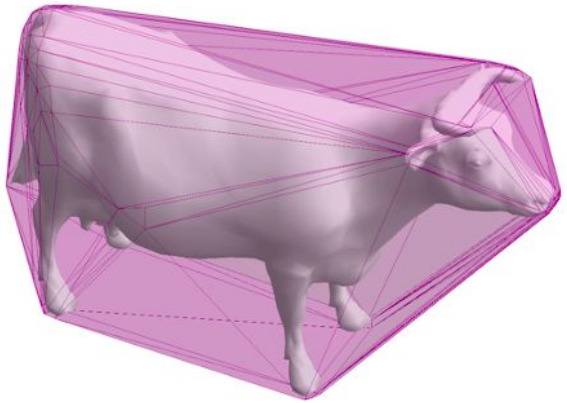


Table 1: Spaces and their corresponding geometric objects derived from a metabolic model and its constraints.

Space	Constraints	Geometric Object
Initial flux space	$Sv = 0$	Linear subspace (null space of S)
Flux cone	$Sv = 0$, $v_i \geq 0$ for all irreversible i	Convex polyhedral cone
Feasible flux space	$Sv = 0$, $v_i^{min} \leq v_i \leq v_i^{max}$	Convex polytope
Thermodynamically feasible flux space	$Sv = 0$, $v_{imin} \leq v_i \leq v_{imax}$, if $v_i \neq 0$, then $v_i G_i \leq 0$	Non-convex polytope

Rounding a polytope





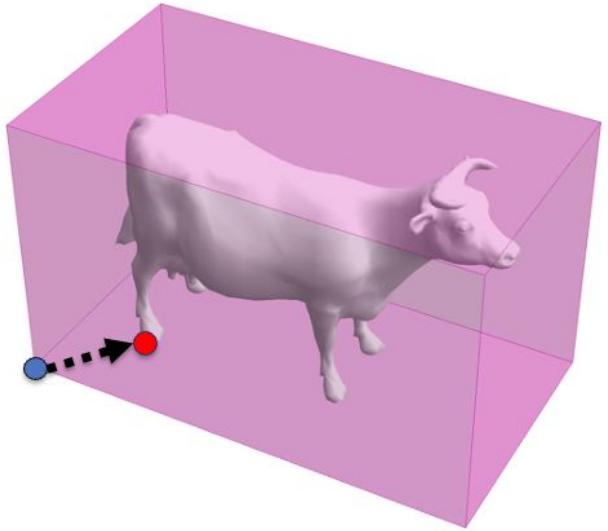


Figure from:
https://nathan-klein.github.io/rounding.html

CHRR

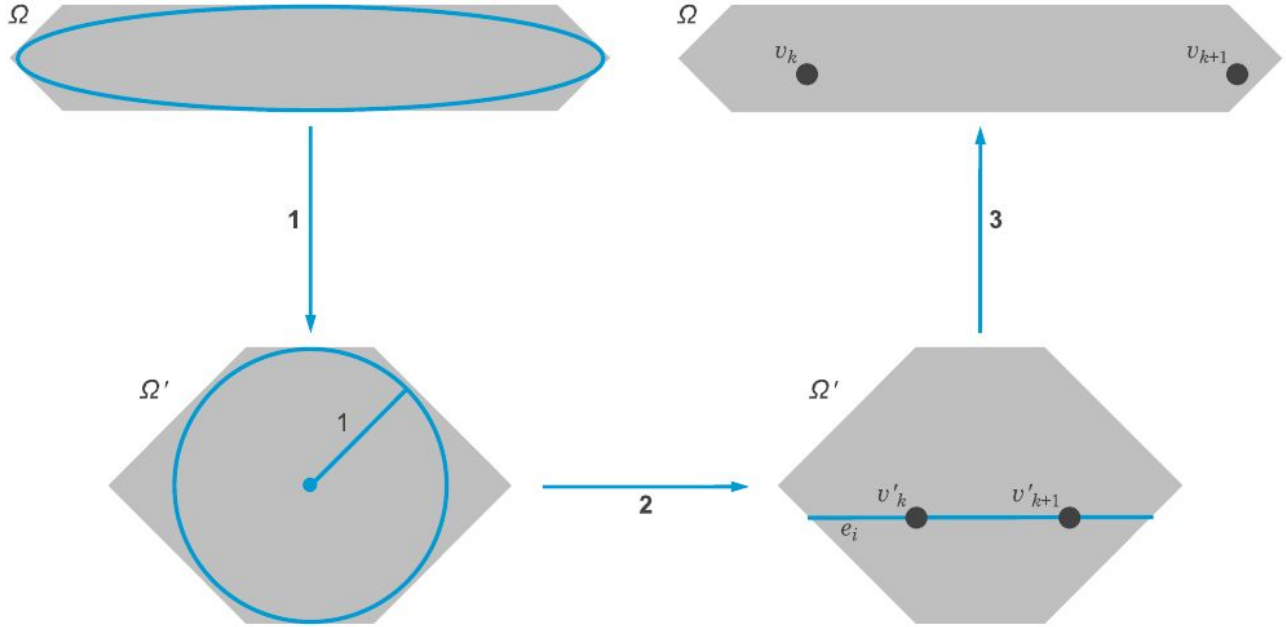


Fig. 1: **Coordinate hit-and-run with rounding.** A high-level illustration of the process to uniformly sample a random metabolic flux vector v from the set Ω of all feasible metabolic fluxes (grey). 1) Apply a rounding transformation T to Ω. The transformed set $\Omega' = T\Omega$ is such that its maximal inscribed ellipsoid (blue) approximates a unit ball. 2) Take q steps of coordinate hit-and-run. At each step, i) pick a random coordinate direction e_i, and ii) move from current point $v'_k \in \Omega'$ to a random point $v'_{k+1} \in \Omega'$ along $v'_k + \alpha e_i \cap \Omega'$. 3) Map samples back to the original space by applying the inverse transformation, e.g., $v_k = T^{-1}v'_k$.

Figure from:
Haraldsdóttir HS, Cousins B, Thiele I, Fleming RM, Vempala S. CHRR: coordinate hit-and-run with rounding for uniform sampling of constraint-based models. Bioinformatics. 2017 Jun 1;33(11):1741-3.

Feature	ACHR	CHRR
Markov property	❌ Non-Markovian (depends on history)	✅ Markovian
Theoretical guarantees	Weak/unproven uniformity	Strong, proven convergence to uniform
Centering	Uses artificial “center” from past samples	Not used; relies on rounding transformation
Mixing efficiency	Often fast in practice, but may bias samples	Slower without rounding, but rounding improves efficiency significantly
Usage in GEMs	Popular historically (fast and easy)	Increasingly preferred in rigorous applications

A Multiphase Monte Carlo Sampling algorithm

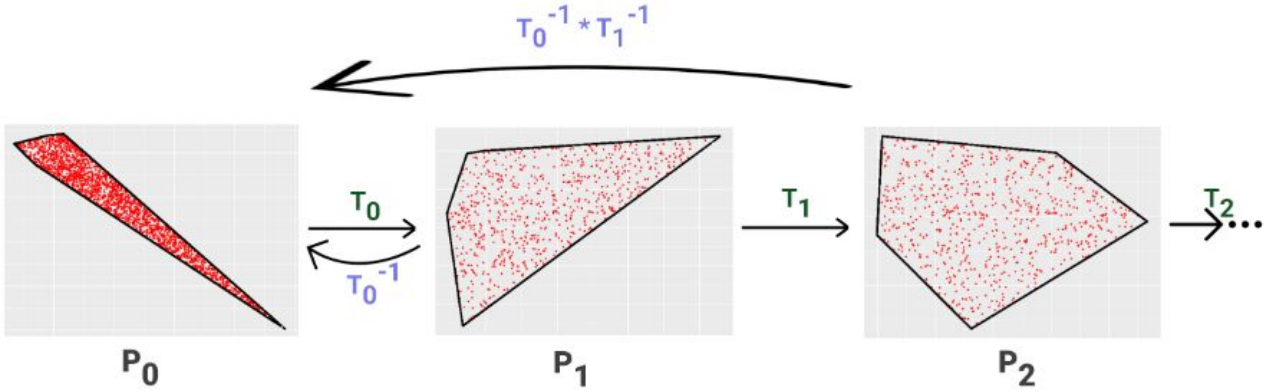


Figure 3 An illustration of our Multiphase Monte Carlo Sampling algorithm. The method is given an integer n and starts at phase $i = 0$ sampling from P_0. In each phase it samples a maximum number of points λ. If the sum of Effective Sample Size in each phase becomes larger than n before the total number of samples in P_i reaches λ then the algorithm terminates. Otherwise, we proceed to a new phase. We map back to P_0 all the generated samples of each phase.

From:
Chalkis A, Fisikopoulos V, Tsigaridas E, Zafeiropoulos H. Geometric algorithms for sampling the flux space of metabolic networks. arXiv preprint arXiv:2012.05503. 2020 Dec 10.

dingo : a Python library for flux sampling

sampling from the flux space of seven GEMs corresponding to polytopes of dimension ranging from 122 to 5335, under the same statistical guarantees.

(the last one is Recon3D Clemence mentioned earlier for being the largest one so far)

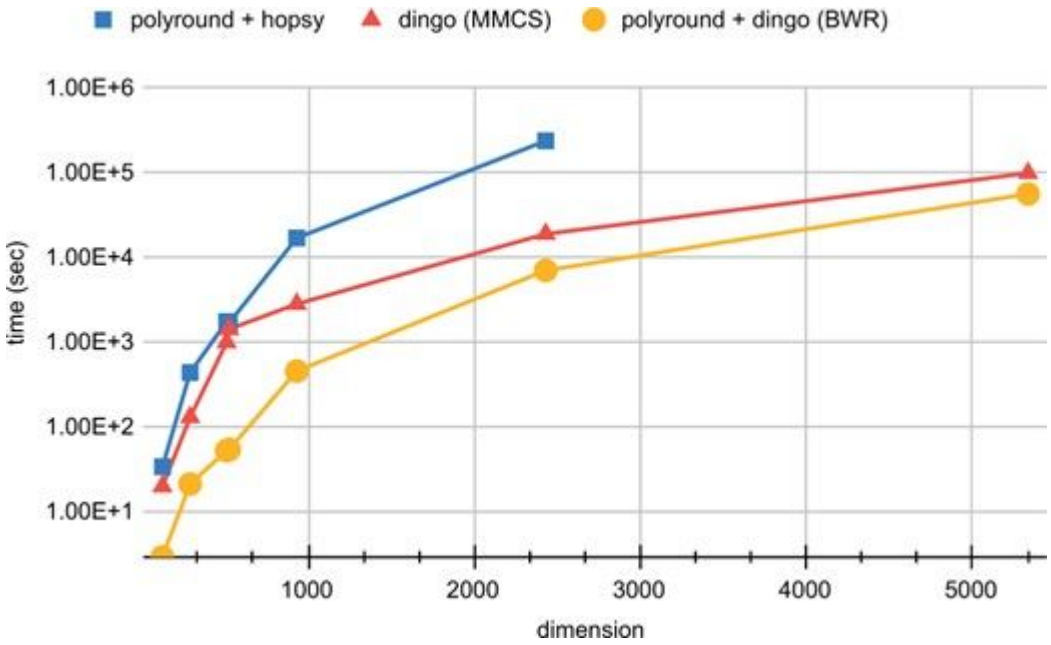


Figure from:
Chalkis A, Fisikopoulos V, Tsigaridas E, Zafeiropoulos H. dingo: a Python package for metabolic flux sampling. Bioinformatics Advances. 2024 Jan 1;4(1):vbae037.

Is uniform the best way to go?

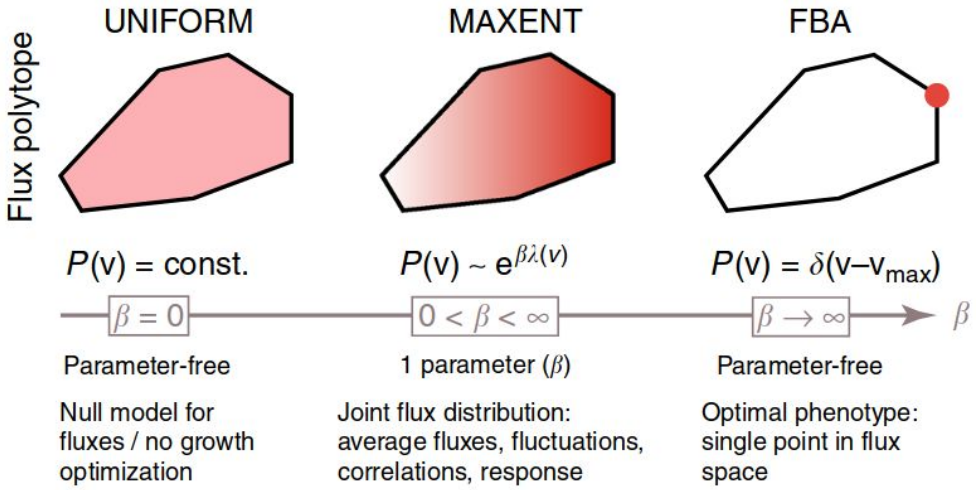
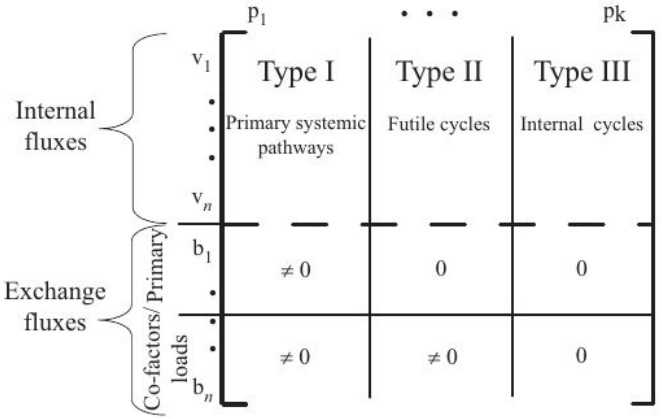


Fig. 1 Stoichiometric and thermodynamic constraints define a high-dimensional convex polytope of permissible fluxes, here shown in cross-section schematically as a black polygon. In a uniform model, the flux distribution, $P(\mathbf{v})$, is uniform over this polytope (UNIFORM). In contrast, flux balance analysis (FBA) finds a single permissible and optimal combination of fluxes, $\mathbf{v}_{max}$ (red dot), such that the growth rate is maximal,

Figure from:
De Martino D, Mc Andersson A, Bergmiller T, Guet CC, Tkačik G. Statistical mechanics for metabolic networks during steady state growth. Nature communications. 2018 Jul 30;9(1):2988.

Loops: may even allow the generation of infinite ATP



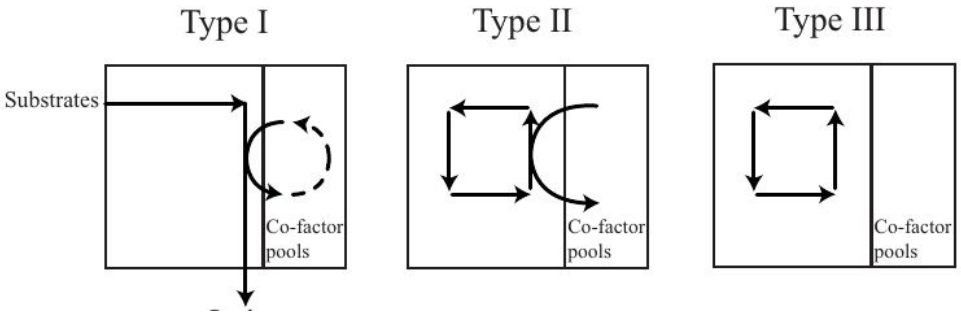


Figure from:
Bernhard Ø. Palsson - Systems Biology_ Constraint-Based Reconstruction and Analysis (2015, Cambridge University Press)

Loopless ? Non-convex

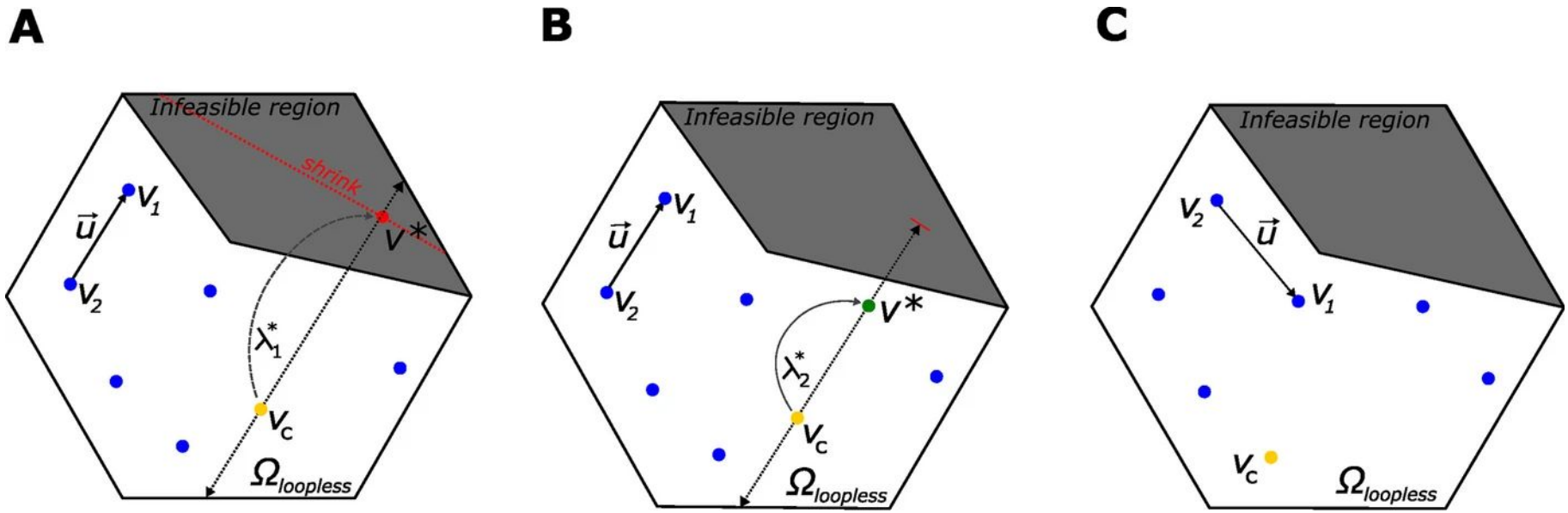


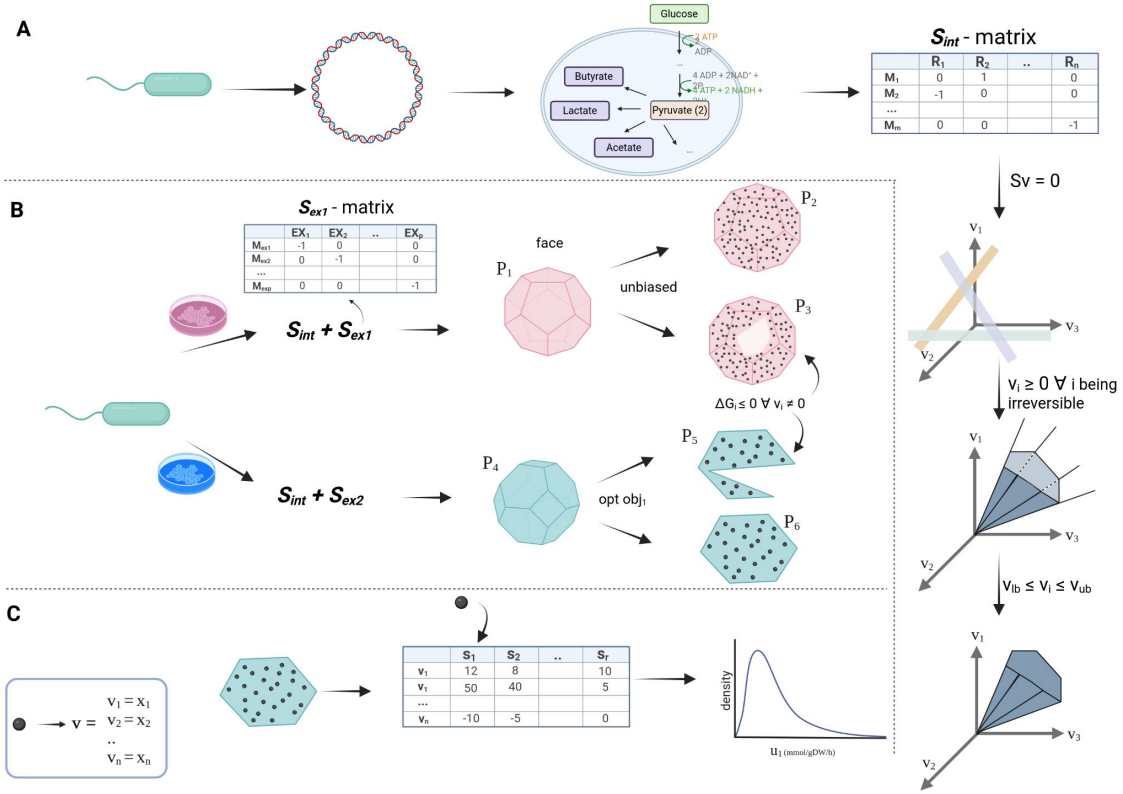
Figure from:
Saa PA, Zapararte S, Drovandi CC, Nielsen LK. LooplessFluxSampler: an efficient toolbox for sampling the loopless flux solution space of metabolic models. BMC bioinformatics. 2024 Jan 2;25(1):3.

MCMC Convergence Diagnostics and quality

- **Effective Sample Size (ESS)**. It is the number of effectively independent draws from the target distribution that the Markov chain is equivalent to. For autocorrelated samples, ESS bounds the uncertainty in estimates [21] and provides information about the quality of the sample.

- **Potential Scale Reduction Factor (PSRF).** an estimated factor by which the scale of the current distribution for the target distribution might be reduced if the simulations were continued for an infinite number of iterations.

An overview of what we have discussed so far!



So, how well is this sampling experiment performing?

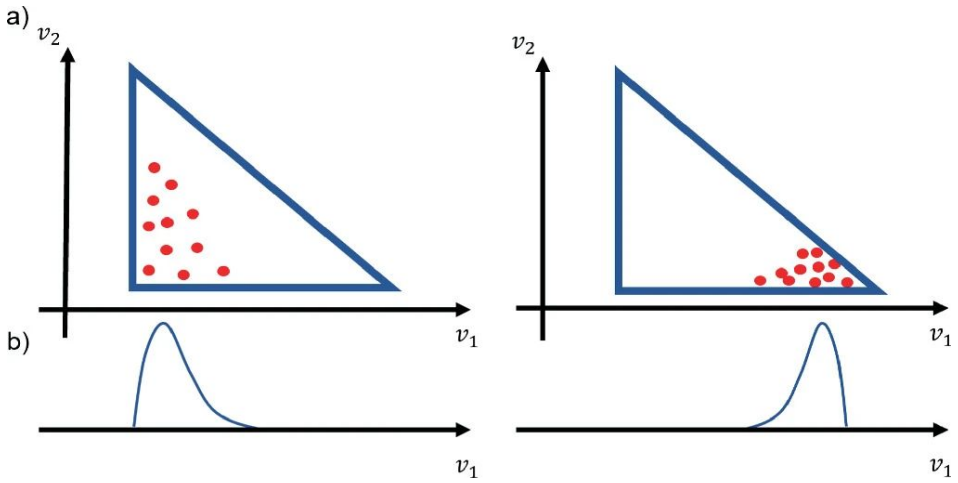


Figure from:
Galuzzi BG, Milazzo L, Damiani C. Best practices in flux sampling of constrained-based models. InInternational Conference on Machine Learning, Optimization, and Data Science 2022 Sep 19 (pp. 234-248). Cham: Springer Nature Switzerland.

Marginal flux distributions of a specific reaction

obtained from two different executions of the same sampling procedure.

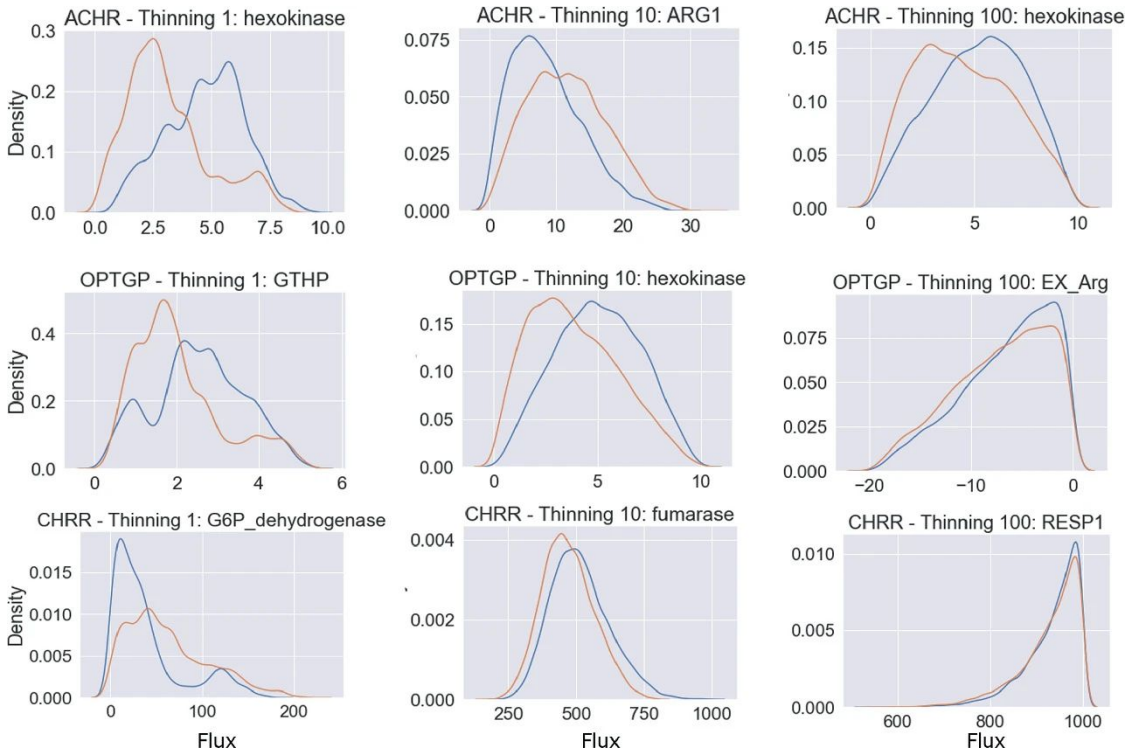
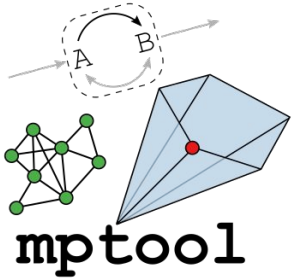
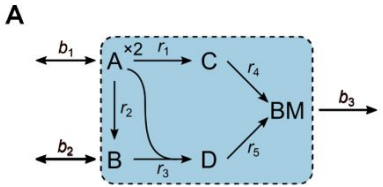


Figure from:
Galuzzi BG, Milazzo L, Damiani C. Best practices in flux sampling of constrained-based models. InInternational Conference on Machine Learning, Optimization, and Data Science 2022 Sep 19 (pp. 234-248). Cham: Springer Nature Switzerland.

Bringing together pathway enumeration and sampling

Minimal pathways (MPs): *defined as minimal sets of reactions in a subnetwork that must have non-zero flux to satisfy all constraints on the full network.*

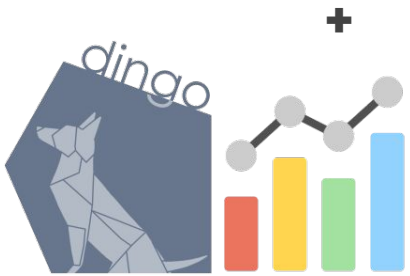


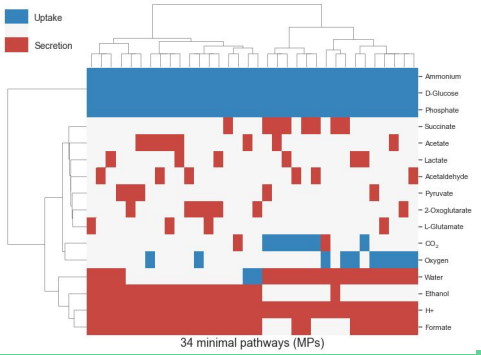


B

	ECM1	ECM2	ECM3	EFP1	EFP2	EFP3	MP1
A	-1	-2	-1	-	-	-	-
B	-1		+1	-		+	
BM	+1	+1		+	+		+

Ove Øyås, Axel Theorell, Jörg Stelling. *Scalable enumeration and sampling of minimal metabolic pathways for organisms and communities*
bioRxiv 2024.10.09.617357; doi: https://doi.org/10.1101/2024.10.09.617357





illustrative example using toy dataset
Figure from: https://gitlab.com/YlvaKaW/exchange-enumeration

Hands-on!

Part A

After a time-out



you have to do better!

So, we play “split”



People experienced in Python, raise your hand, and “move the ball”.

Call for open-source

A lot of software depend on commercial solvers.

For example, carveme needs either CPLEX or Gurobi

Solver

You also need an optimization solver. The recommended option is to install either IBM CPLEX or Gurobi. You need a full version with the respective academic license (demo version will not work).









Let’s jump to the notebook

each of you, has a unique codespace, thus a unique link to your notebook

The first task for you and your team

- Choose a species
- Solve an FBA
- Choose a sampler and a solver and sample
- Compare uptake and secretion vectors of FBA and your sampling

bh_model.metabolites[0]

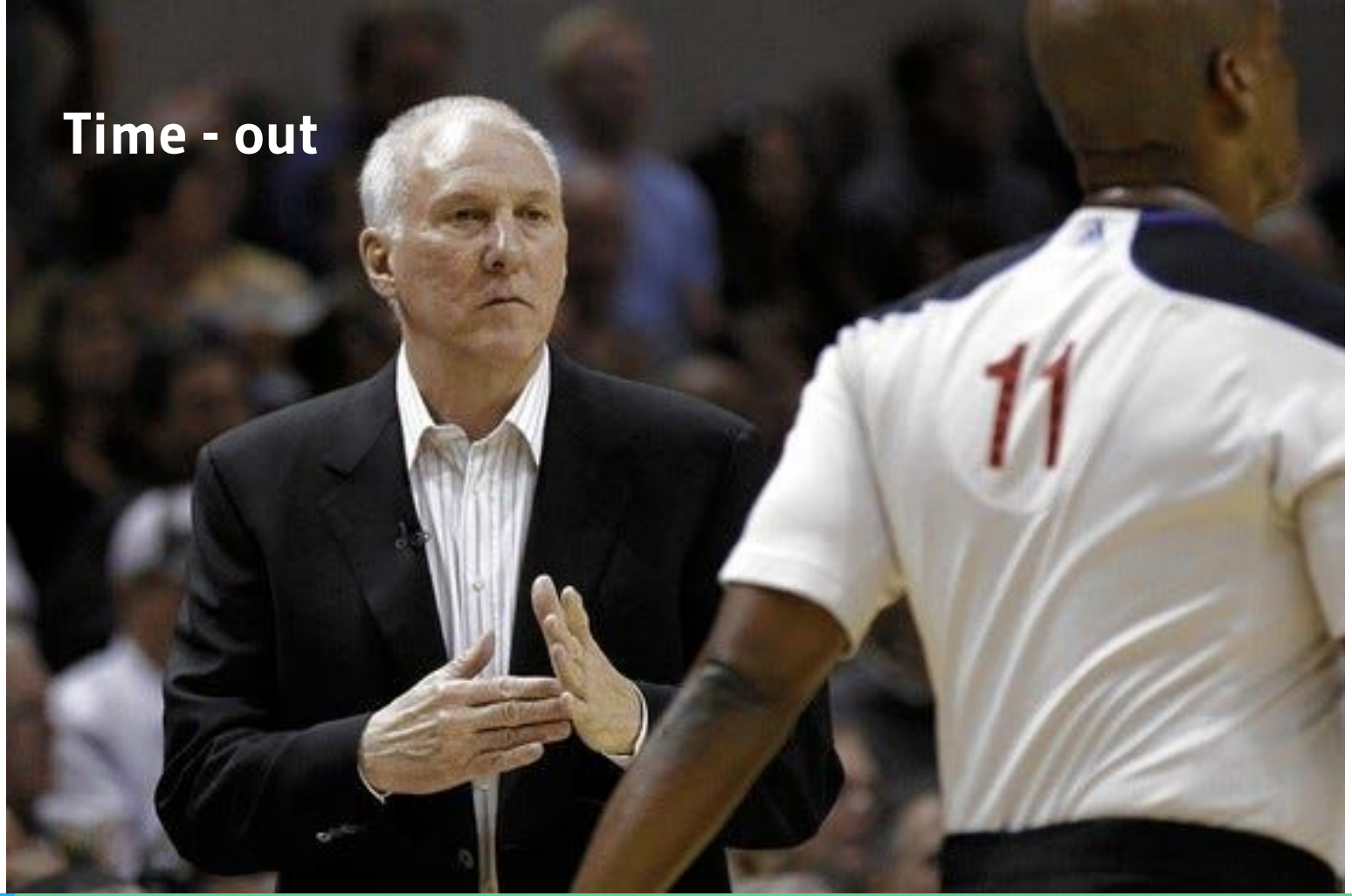
If that was easy for your team, then..

- Create a copy of your model, and then

A. **Case 1**
   a. **Edit your media** by increasing/decreasing sets of exchange reactions bounds
   b. Sample again and compare with the original case

B. **Case 2**
   a. Use `dingo` to sample sub-optimal cases and compare with original case

Time - out



Hands-on!

Part B

Monte Carlo sampling is not itself machine learning

but it is a **general computational technique** that is often used **inside machine learning** and other fields (statistics, physics, finance, etc.).

- **Monte Carlo sampling**: A family of methods that use repeated random sampling to approximate expectations, integrals, or probability distributions. It's a numerical tool, not a learning algorithm.
- **Machine learning**: The field focused on building models that learn patterns or make predictions from data.

👉 Where they meet among other tasks:

- Monte Carlo is often used in **Bayesian machine learning** (e.g., Markov Chain Monte Carlo for sampling posterior distributions).
- It's used in **reinforcement learning** (Monte Carlo methods estimate value functions by averaging returns)

The findings of the sampling though,
quite a descent dataset to apply ML to

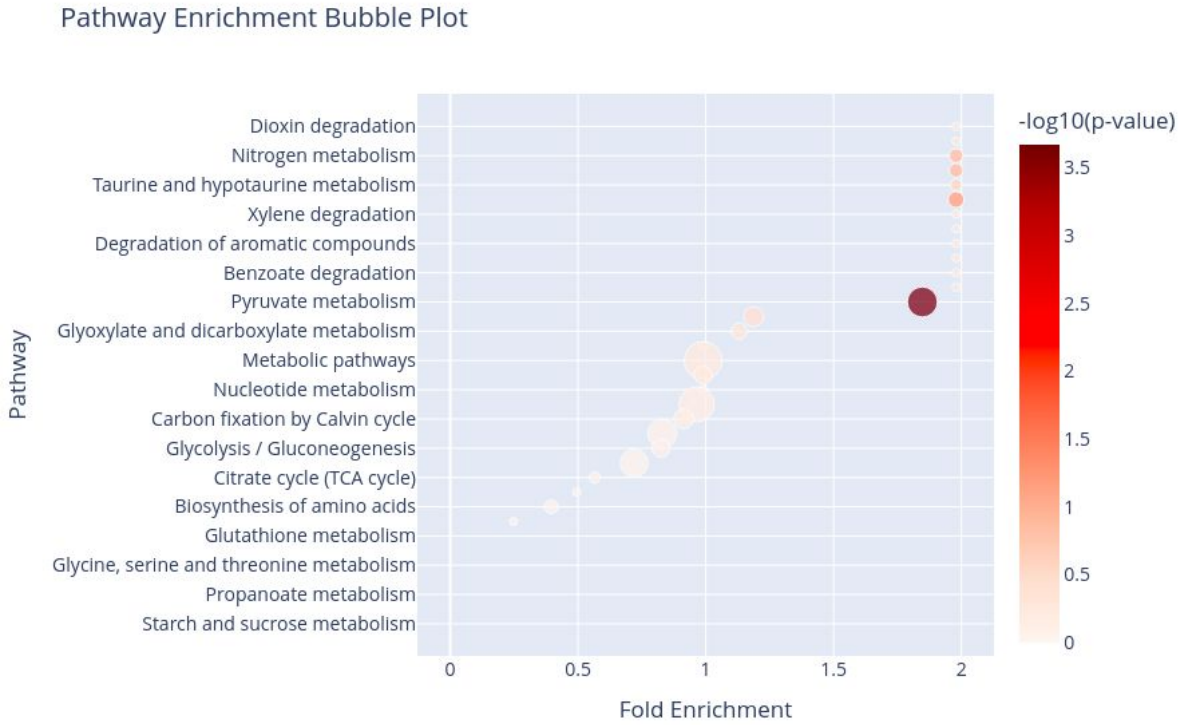
https://dingo-stats.readthedocs.io/



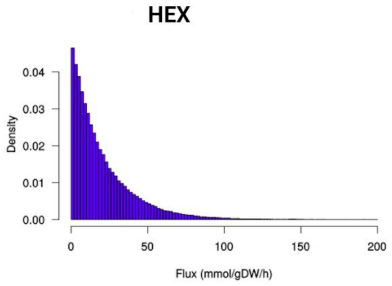
Here is your final task!

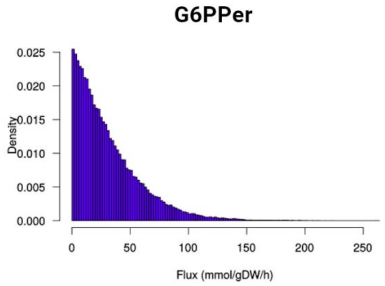
- Choose a species
- Calculate its Minimal Pathways (MPs) using `mptool`
- Fig

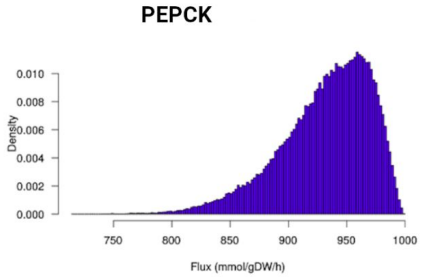
Remember what Julian Tap mentioned about significance?

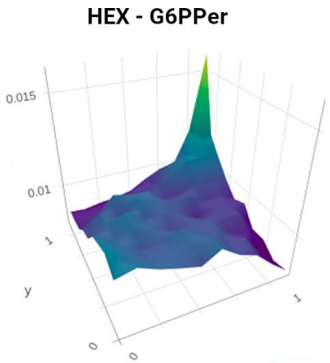


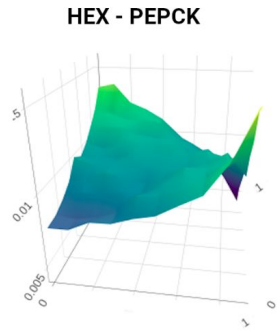
Remember what Alex Fedorec mentioned about visualizing variables changing together?











A **copula** is a bivariate probability distribution for which the marginal probability distribution of each variable is uniform.

It captures the dependency between two fluxes.

HEX:
atp_c + **glc__D_c** → adp_c + g6p_c + h_c :

G6PPer:
g6p_r + h2o_r → pi_r + **glc__D_r**

PEPCK:
gtp_c + oaa_c → co2_c + gdp_c + pep_c

Constraining a GEM

“If you don’t look at where you want to go, you’ll end up where you are looking”

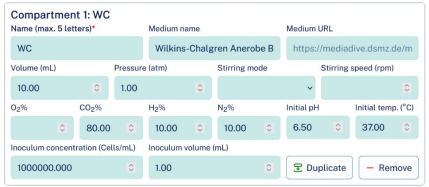
Querying, visualizing, and sharing microbial growth curve data

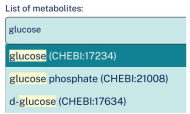


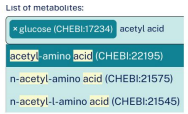
Andrew Radev

µGrowthDB
Microbial Growth Database

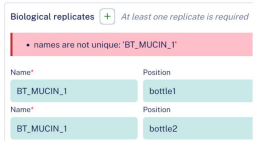


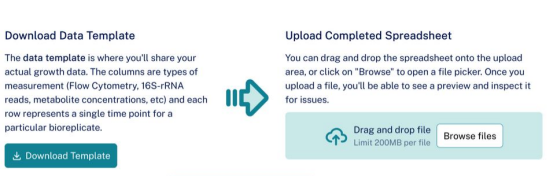










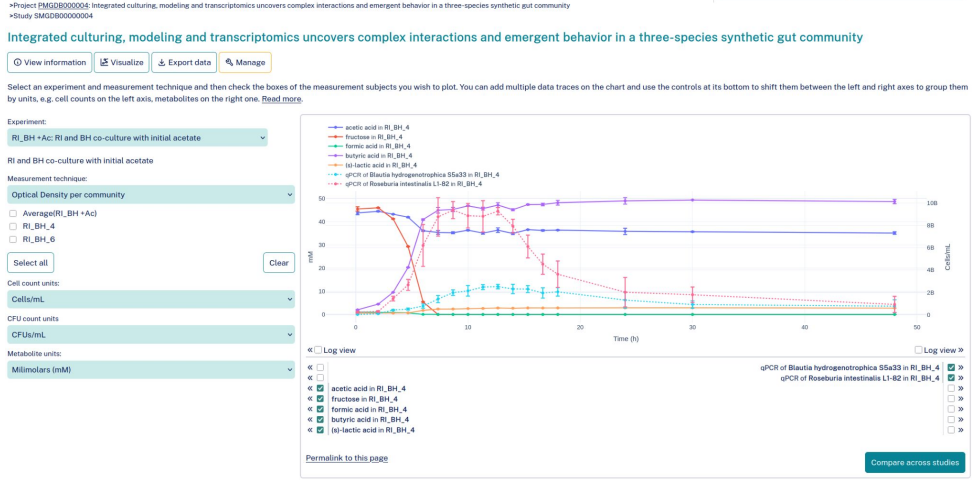


template_data_filled.xlsx 58.0 kB | Growth data per community

Biological Replicate	Compartment	Time	Community OD	Community pH
BT_MUCIN_1	WC	0	0.012	6.64
BT_MUCIN_1	WC	4	0.034	6.64

	A	B	C	D	E
1	Biological Replicate	Compartment	Time	Roseburia intestinalis FC counts	Bacteroides thetaiotaomicron FC counts
278	RI_WC_3	WC	38	266115	
279	RI_WC_3	WC	48	381730	
280	RI_WC_3	WC	60	337310	
281	RI_WC_3	WC	72	240980	
282	RI_WC_3	WC	96	305040	
283	RI_WC_3	WC	120	225940	
284	BTRI_MUCIN_1	WC	0	747	1025
285	BTRI_MUCIN_1	WC	4	5339	14880
286	BTRI_MUCIN_1	WC	8	90780	316120
287	BTRI_MUCIN_1	WC	12	149020	1024600
288	BTRI_MUCIN_1	WC	16	212940	1762500
289	BTRI_MUCIN_1	WC	20	114320	1577600
290	BTRI_MUCIN_1	WC	24	155760	1219200
291	BTRI_MUCIN_1	WC	28	173560	1360040
292	BTRI_MUCIN_1	WC	32	117380	1056120
293	BTRI_MUCIN_1	WC	36	81460	998140
294	BTRI_MUCIN_1	WC	40	42720	811320
295	BTRI_MUCIN_1	WC	48	11300	959380
296	BTRI_MUCIN_1	WC	60	5200	915600
297	BTRI_MUCIN_1	WC	72	1460	1010220
298	BTRI_MUCIN_1	WC	96	1300	1007920
299	BTRI_MUCIN_1	WC	120	3300	986600
300	BTRI_MUCIN_1	MUCIN	0	0	0
301	BTRI_MUCIN_1	MUCIN	4	0	8700
302	BTRI_MUCIN_1	MUCIN	8	0	105900
303	BTRI_MUCIN_1	MUCIN	12	2750	130750
304	BTRI_MUCIN_1	MUCIN	16	26500	340200
305	BTRI_MUCIN_1	MUCIN	20	60500	715750

Growth data per community | Growth data per strain | Growth data per metabolite



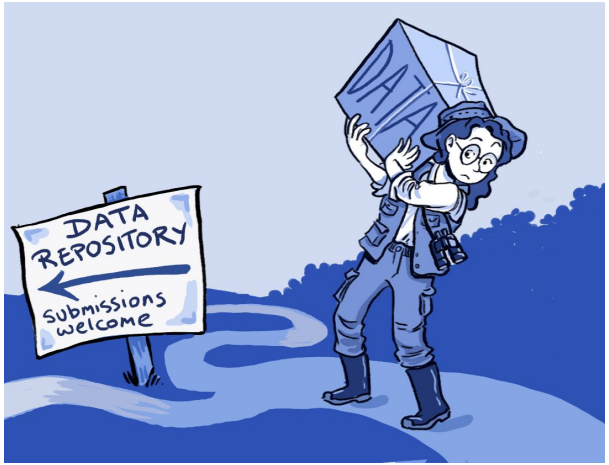
https://mgrowthdb.gbiomed.kuleuven.be/

Add value to your data

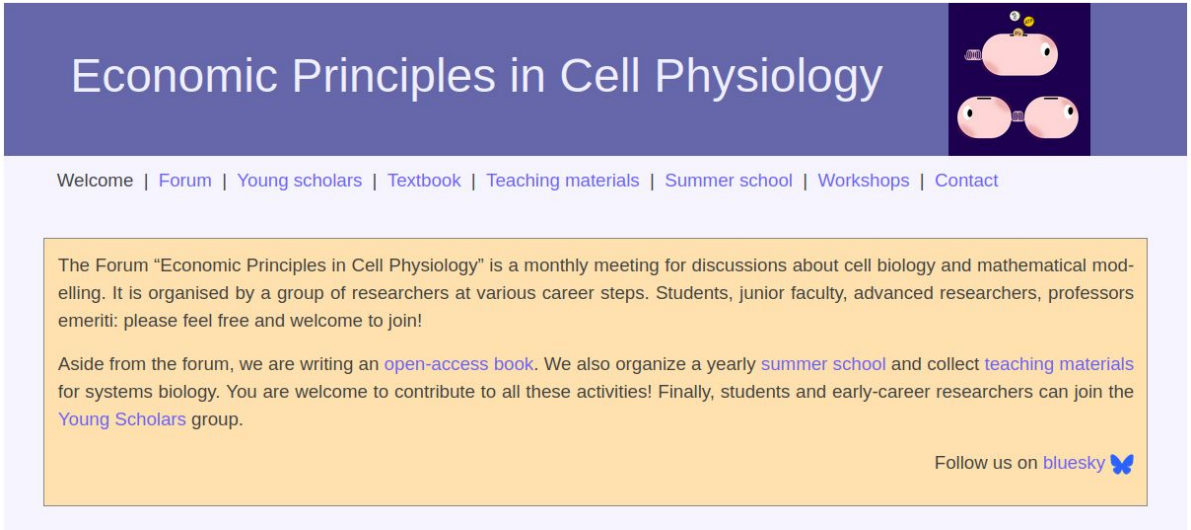
GEMs refinement → more accurate predictions

but, not only GEMs benefit

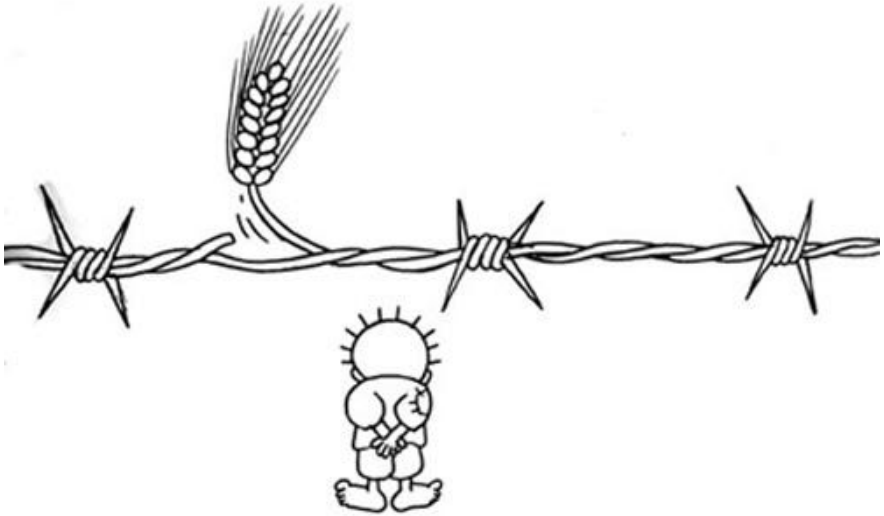
10:30 - 12:30	Simulation and control of high-dimensional dynamical systems using artificial neural networks	Lucas Böttcher



If you are interested in today's talks, keep in mind EPCP



https://principlescellphysiology.org/





Thank you for your attention!



Karoline Faust



Daniel R. Garza

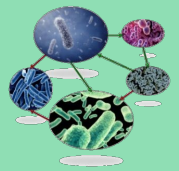


Sotiris Touliopoulos



Apostolos Chalkis

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Google
Summer of Code